HF-PULSE: Design and Pre-Specified Evaluation of a Stateful, Wearable-Driven Digital Twin for Heart Failure Home Monitoring

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***Abstract*—Heart-failure (HF) patients monitored at home send daily measurements, but nothing explains what those numbers mean for the body. We built HF-PULSE, an open-source model that turns daily wearable data into a simulated picture of the heart and circulation. A machine-learning model reads three weeks of six daily signals plus routine clinical values and estimates what kind of decline may be under way and how severe it is. That estimate drives the Pulse physiology engine. The engine's simulated heart and blood-pressure values feed a readable risk score with three alert levels. The model can also run day after day, carrying its state forward, and a paused-and-resumed run gives the same results as an uninterrupted one. We tested HF-PULSE on ten scripted 21-day patient stories and wrote down the expected results before any run. A check built on the risk score cut false alerts by 61–68% without missing any declining patient. We then re-ran the tests with patients' heart-pumping strength (ejection fraction) inside the range the model was trained on. Day-by-day accuracy doubled (17.6% to 37.1%), 24 of 30 declining cases were caught (up from 21), and false alerts nearly halved. Seven of our ten written predictions came true. However, the physiology engine did not catch more cases than the machine-learning model alone. What it adds is an explanation of each alert, and whether that helps clinicians is still untested. HF-PULSE also ran on real home data from 13 of 16 HF patients. On 17,129 hospital admissions (MIMIC-IV), the only part that works without wearables did about as well as age alone (AUC 0.596).**

***Index Terms*—Digital twin, heart failure, home monitoring, wearable sensors, physiology simulation, Pulse engine, random forest, remote patient monitoring, early warning, pre-specified evaluation, reproducibility.**

I. Introduction

A. Health challenge

Heart failure affected an estimated 64.3 million people worldwide in 2017 [1], [2] and is defined by progression: patients decline over days to weeks rather than instantaneously. Current guidelines emphasize early recognition of congestion and decompensation [3].

Yet three structural gaps remain in practice. First, monitoring is episodic: patients are typically seen every few weeks, and a patient can move from stable to crisis between visits. Second, consumer wearables report numbers, not physiology: a rising resting heart rate is measured, but whether it reflects benign variation or an early compensatory response to falling cardiac output is left to the patient. Third, threshold alarms (e.g., $\mathrm{SpO}_2 < 92\%$) fire only after a value is already abnormal; they describe where a patient is, not where they are heading.

B. The digital-twin approach

Remote monitoring has shown benefit when it captures physiology rather than symptoms alone, from structured telemonitoring [4] to implanted pulmonary-artery pressure sensors [5], and multi-sensor wearables combined with machine learning have predicted HF hospitalization days in advance [6], [7]. These data-driven predictors, however, offer little mechanistic explanation of *why* an alert fired.

Cardiac digital twins promise exactly that mechanistic layer [8]–[10], but most are built from imaging or invasive measurements fitted to one patient at a time [11], [12], and few are evaluated over weeks of everyday data.

We report a case study of a twin that sits between these two traditions: it takes daily home-monitoring trends (wearable and home-device signals) as input, interprets them through a mechanistic whole-body physiology engine (Kitware Pulse [13]), and produces interpretable risk, staging and forward projections. We use "twin" in a limited sense: population statistics parameterize the engine body, and the patient's ejection fraction and estimated severity scale it; we do not fit it to the individual.

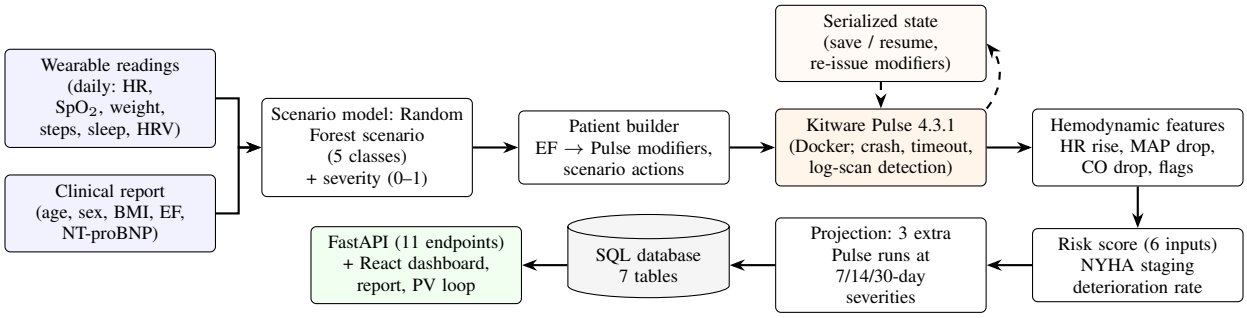


Fig. 1. System architecture. A filled 21-day wearable window triggers one background assessment; the dashed loop is the continuous-state mode, in which each day's simulation resumes from the previous day's serialized engine state (Section IV).

C. Research methodology

Fig. 1 shows the pipeline. A patient record holds demographics and a clinical report (ejection fraction, NT-proBNP); daily wearable readings accumulate until they fill a 21-day window, which triggers one background assessment: (1) the scenario classifier and severity regressor (the "scenario model"); (2) construction of a Pulse patient and scenario file; (3) simulation inside the Pulse container with crash and timeout detection; (4) hemodynamic feature extraction; (5) risk scoring, NYHA staging and trend-based deterioration rate; and (6) three further Pulse runs at projected severities for 7, 14 and 30 days. Section III details each stage, Section IV the engine problems we solved, Section V the evaluation on synthetic patients and Section VI the tests against real data.

D. Contributions

Our contributions are:

1) A complete, open-source pipeline (wearable trends → ML scenario and severity estimation → Pulse simulation → interpretable risk scoring, staging and projection), deployed as a service with a dashboard and an optional continuous mode that carries engine state from day to day, with resumed outputs matching uninterrupted runs (Sections I-C and III).
2) Engine-level findings for using Pulse as an HF twin: mapping ejection fraction onto the engine, a root-caused instability boundary when disease modifiers and exertion combine, a state save/resume method whose outputs match uninterrupted runs (Section IV), and a baroreflex set-point reset that the saved state carries from day to day (Section V-H).
3) A pre-specified, replicated evaluation (Sections V and VI): a 21-day scenario test with development and held-out seeds, a replication on new seeds after a predicted fix, an ablation against the classifier alone, a label-only rule and a clinical weight rule, and real-patient and real-outcome tests. It shows what the physiology layer adds (an alert guard built on its hemodynamic components that cut false alerts by 61–68%, and a mechanistic account of every alert), shows that every high-risk day followed an exertion label that switched on simulated exercise, and shows that a tuned classifier detected more deteriorating series at the same false-alert rate.
4) A reproducible research artifact: frozen, hash-checked models, 353 automated test cases passing in continuous integration, research features behind default-off flags with a golden-file parity test, and a fresh-clone reproduction of the demonstration patients.

II. Literature Survey

We group prior work into five themes: digital twins in healthcare and cardiology, physiology engines and lumped cardiovascular models, wearable and implanted HF monitoring, HF risk scores and machine learning, and the validation of digital twins. We then position HF-PULSE against them and state the gaps we address.

A. Digital twins in healthcare and cardiology

Venkatesh *et al.* [14] define a health digital twin as a virtual representation of a patient generated from multimodal patient data, population data and real-time updates, and name computation, clinical implementation, data governance and product oversight as the main considerations for translation. Balasubramanyam *et al.* [15] surveyed the 2020–2023 literature on digital twins in healthcare: their roles and benefits, the layers needed to implement a twin-based smart healthcare system, and its limitations and challenges. Katsoulakis *et al.* [16] concluded in a scoping review that digital twins for health are still at an early stage.

In cardiology, Corral-Acero *et al.* [8] framed the digital twin as the route to precision cardiology, Coorey *et al.* [9] reviewed the emerging health-twin field for cardiovascular disease, and Niederer *et al.* [10] argued that twins must move from artisanal, single-patient builds to scalable, industrial pipelines.

Recent HF-specific twins fit mechanistic cardiovascular models to measured data. Gu *et al.* [11] fitted twins to 343 HF patients and used the fitted parameters for interpretable phenotyping and prognosis. Geddes *et al.* [12] built CFD-based twins that non-invasively estimate markers of right HF. Balasubramanyam *et al.* [17] proposed a cardiovascular twin combining time-varying compliance, Fourier-based ECG reconstruction and 3D anatomical modeling.

B. Physiology engines and lumped cardiovascular models

Zero-dimensional (lumped-parameter) models represent the components of the cardiovascular system in simplified, lumped form. Shi *et al.* [18] reviewed them as a concise way to evaluate hemodynamic interactions among cardiovascular organs, with one-dimensional models adding pulse-wave transmission at far lower cost than three-dimensional CFD. The arterial Windkessel is the classic lumped element: Frank's two-element form, a resistance and a compliance, explains diastolic pressure decay, and a third element, the characteristic impedance, improves its behavior in systole; as a lumped model it cannot represent spatially distributed phenomena, but it approximates ventricular afterload simply and fairly accurately [19]. For the heart itself, Suga and Sagawa [20] showed in the excised canine left ventricle that the instantaneous pressure–volume relationship is approximated by a time-varying elastance, $P(t) = E(t)\,[V(t) - V_d]$, the concept on which the elastance models in Section III-D build.

Three open-source engines package such models at whole-body or whole-circulation scale. Pulse [13], [21] and BioGears [22] both model physiology with electric-circuit analogues; BioGears targets medical education, research and training and supports in-silico trials [22]. CircAdapt [23] simulates the beat-to-beat dynamics of the four-chamber heart with systemic and pulmonary circulations, and lets cardiac and vascular tissue adapt to mechanical load through local rules. We build on Pulse, whose open-source engine accepts actions and conditions during a run and can serialize and resume its full state (Section IV).

C. Wearable and implanted HF monitoring

Non-invasive telemonitoring trials have mixed results. Tele-HF [24] randomized 1,653 recently hospitalized patients to daily telephone-reported symptoms and weight or usual care and found no significant difference in readmission or death within 180 days (52.3% vs. 51.5%). BEAT-HF [25] combined health-coaching calls with telemonitoring of blood pressure, heart rate, symptoms and weight in 1,437 patients and did not reduce 180-day readmissions (50.8% vs. 49.2%; adjusted HR 1.03). TIM-HF2 [26], a structured remote patient management program in 1,571 patients, reduced the percentage of days lost to unplanned cardiovascular admission or all-cause death (4.88% vs. 6.64%; ratio 0.80, 95% CI 0.65–1.00) and all-cause mortality (HR 0.70). Earlier structured telephone support and telemonitoring programs were reviewed by Inglis *et al.* [4].

Implanted sensors measure physiology more directly. In CHAMPION [5], management guided by a wireless pulmonary-artery pressure sensor lowered the 6-month rate of HF hospitalization (0.31 vs. 0.44; HR 0.70). GUIDE-HF [27] extended this approach to NYHA class II–IV patients, and its primary endpoint, a composite of mortality and total HF events at 12 months, was not significantly reduced (HR 0.88, 95% CI 0.74–1.05). The HeartLogic index from the MultiSENSE study [28] combines heart sounds, respiration, thoracic impedance, heart rate and activity from implanted CRT defibrillators; in its sequestered test cohort it detected 70% of HF events with 1.47 unexplained alerts per patient-year and a median lead time of 34 days. MultiSENSE reported unexplained alerts per patient-year, the device-study analogue of our false alerts per 100 patient-days.

Wearable predictors aim for similar performance without an implant. The LINK-HF study [6] used a disposable multisensor chest patch and a machine-learned personalized baseline: deviations from the baseline detected precursors of HF re-hospitalization with 76–88% sensitivity and 85% specificity, and the median time from first alert to readmission was 6.5 days. Gautam *et al.* [7] surveyed AI, wearables and remote monitoring in HF.

D. HF risk scores and machine learning

The Seattle Heart Failure Model [29] was derived in 1,125 patients with a Cox model and validated in five cohorts totalling 9,942 patients, with an overall AUC of 0.729 for survival. The MAGGIC score [30] is a widely used, externally validated HF survival score derived from 39,372 patients in 30 studies, and the LACE index [31] is a general, not HF-specific, score for early death or unplanned readmission after discharge. We use MAGGIC as an external benchmark (Section VI-C). Across 117 HF prediction models in 55 papers, Ouwerkerk *et al.* [32] found a mean C-statistic of 0.66 (0.71 for mortality, 0.68 for HF hospitalization) and judged prediction, especially of HF hospitalization, only moderately successful.

Machine learning has improved these figures modestly. On the Tele-HF trial data, Mortazavi *et al.* [33] found that random forests raised the C-statistic for 30-day all-cause readmission from 0.533 (logistic regression) to 0.628, and boosting raised it for HF readmission from 0.543 to 0.678. Chicco and Jurman [34] predicted survival from serum creatinine and ejection fraction alone. None of these models simulates the physiology that produces the risk.

E. Validating digital twins

A twin's predictions need evidence of credibility before they inform care. Viceconti *et al.* [35] describe a framework for assessing the credibility of mechanistic ("biophysical") models used in the regulatory evaluation of medical products, built on verification, validation and uncertainty quantification and beginning with the definition of the model's context of use. Pathmanathan and Gray [36] note that credibility is established primarily by validation against experimental or clinical data, but that for complex multiscale models it often rests on many distinct kinds of evidence, so the extent to which such a model can be trusted is often unclear.

Pre-specification guards the evaluation itself. Nosek *et al.* [37] argue that defining the research questions and analysis plan before observing outcomes separates predictions from postdictions, and describe strategies for cases that fall short of the ideal, such as preexisting data. Our scenario tests follow this logic with outcomes committed to version control before any run, which is weaker than registration in a public registry (Section V-D).

TABLE I
POSITIONING AGAINST RELATED WORK

Work	Input data	Model	Personalization	Gap addressed here
Corral-Acero 2020 [8], Coorey 2022 [9], Balasubramanyam 2024 [15]	Multimodal (vision/review)	Mechanistic + statistical	Conceptual	No deployed system or longitudinal test
Gu 2025 [11]	Clinical, echo, invasive	Lumped cardiovascular model + unsupervised ML	Fitted per patient (343 HF)	Requires hospital measurements; no home monitoring
Geddes 2025 [12]	Imaging, invasive validation	3D CFD	Fitted per patient	Right HF only; not wearable-driven
Balasubramanyam 2025 [17]	ECG datasets, echo	Compliance model, Fourier ECG, 3D mesh	Signal-level	Visualization and ECG fidelity, not deterioration tracking
Boehmer 2017 (HeartLogic) [28]	Implanted CRT-D sensors	Multisensor composite index	Device-level	Requires an implanted device; no physiological simulation
Stehlik 2020 (LINK-HF) [6]	Wearable patch	ML (similarity-based)	Patient baseline	No mechanistic explanation of alerts
This work	Wearable trends + routine clinical	RF + Pulse engine + interpretable score	Scenario/severity-driven, EF-scaled	Mechanistic, wearable-driven, stateful; evaluated over 21 days with pre-specified outcomes

F. Positioning and gaps addressed

Table I contrasts these approaches. Mechanistic twins personalize deeply but require hospital measurements; wearable predictors scale but are opaque. Our twin trades per-patient parameter fitting for deployability from home data, and adds a mechanistic layer that wearable-only predictors lack. We claim novelty for the integrated pipeline and its evaluation, not for any single component. Specifically, we address four gaps:

1) *Input data.* The HF twins in Section II-A are built from imaging, echocardiography or invasive measurements; we drive a mechanistic engine from daily wearable trends and routine clinical values.
2) *Explanation.* Wearable and implanted predictors (Section II-C) output an index or an alert; we attach a hemodynamic breakdown to each alert, whose clinical usefulness remains untested.
3) *State.* We carry the engine's physiological state from day to day instead of rebuilding the patient for each assessment (Section IV).
4) *Evaluation.* We evaluate over 21-day trajectories against outcomes committed before the runs, on development and held-out seeds and a replication, and report false-alert burden per patient-day as device studies do.

III. METHODS

A. Reference data and synthetic training population

We used no real patient data for training. Instead, we took the generator's population statistics from aggregate statistics of real sources (Table III): 11,837 MIMIC-IV HF admissions [38] for age and sex (the extract also gives laboratory reference values, which the generator does not yet use); the Chicco–Jurman HF clinical records [34], the only source with ejection fraction, for the HFrEF and HFpEF profiles; NHANES for height, weight and BMI by sex; and published age-specific NT-proBNP cut-offs [39]. The remaining parameters are documented assumptions: the healthy EF profile, the wearable baselines, the scenario changes $\Delta_{v,c}$ (Table II) and the coefficients that link EF, severity and NT-proBNP. The repository stores only aggregate statistics from the real sources; raw files remain outside version control.

We generated $n = 2{,}000$ synthetic patients across five scenarios. Each patient receives a scenario c and a severity

$$\sigma = \begin{cases} 0.15\,\xi, & c = \text{stable}, \\ \xi, & \text{otherwise}, \end{cases} \qquad \xi \sim U(0,1), \tag{1}$$

so stable patients' severity is scaled into $[0,\ 0.15]$.

1) Clinical state: The generator draws ejection fraction from the scenario's EF profile $p(c)$ (healthy for stable; HFpEF for deconditioning and cardiac stress; HFrEF for fluid overload and acute deterioration) and lowers it by 12σ points:

$$\text{EF} = \text{clip}\big(\mathcal{N}(\mu_p,\ (w_p/4)^2) - 12\sigma,\ 15,\ 75\big), \tag{2}$$

where μ_p and w_p are the profile's mean and range width (HFrEF: 32.3%, range 14–40%; HFpEF: 56.9%, range 50–80%; healthy: 62%, range 55–70%). A mix of EF deficit and severity, scaled to age-specific diagnostic cut-offs, drives NT-proBNP, so BNP falls with EF and rises with severity:

$$d_{\text{EF}} = \text{clip}\big((70 - \text{EF})/55,\ 0,\ 1\big), \tag{3}$$

$$b = \text{clip}\big(0.6\,d_{\text{EF}} + 0.4\,\sigma,\ 0,\ 1\big), \tag{4}$$

$$\text{NT-proBNP} = k(a)\,(0.25 + 2b)\,\eta, \qquad \ln\eta \sim \mathcal{N}(0, 0.25^2), \tag{5}$$

where $k(a)$ is the age-specific NT-proBNP cut-off for acute HF: 450, 900 and 1,800 pg/mL for ages below 50, 50–75 and above 75 years [39]. The lognormal factor η keeps BNP positive and right-skewed.

2) Wearable trends: Each wearable signal v on day t then follows

$$v(t) = b_v + \Delta_{v,c}\,\sigma\,g_m(t/20) + \epsilon_{v,t}, \tag{6}$$

where b_v is a patient baseline (drawn from population statistics, or the patient's own weight), $\Delta_{v,c}$ is the maximum change

TABLE II
WEARABLE BASELINES AND MAXIMUM SCENARIO CHANGES $\Delta_{v,c}$ IN (6)

Signal v	**Baseline**	**FO**	**CS**	**DC**	**AD**
Weight (kg)	own weight	$+4.5$	$+0.5$	$+1.0$	$+3.0$
Resting HR (bpm)	70 ± 8	$+12$	$+20$	$+8$	$+25$
SpO_2 (%)	97 ± 1.2	-2.5	-1.5	-0.5	-5.0
Steps/day	6000 ± 2000	-1200	-1000	-2500	-2000
Sleep (h)	7.0 ± 1.0	-0.8	-0.5	-0.3	-1.2
HRV RMSSD (ms)	35 ± 12	-6	-12	-5	-15

FO: fluid overload; CS: cardiac stress; DC: deconditioning; AD: acute deterioration. Baselines are population mean $\pm$ SD. Stable patients have $\Delta_{v,c} = 0$. Source: `generate_wearable_trends.py` and `reference_stats.yaml`.

for scenario c (Table II; e.g., fluid overload: $+4.5$ kg weight, $+12$ bpm resting HR, -2.5 points SpO_2), g_m is a progression curve for mode m,

$$g_m(u) = \begin{cases} u, & \text{deteriorating, } c \neq \text{AD}, \\ u^2, & \text{deteriorating, } c = \text{AD}, \\ \max(0,\, 1 - 1.15u), & \text{recovering}, \\ 0, & \text{stable}, \end{cases} \tag{7}$$

with AD denoting acute deterioration, and $\epsilon_{v,t}$ is independent Gaussian noise with SD equal to 15% of the population SD (0.4 kg for weight). Over the 21-day window $u = t/20$ runs from 0 to 1, so a deteriorating patient reaches the full change $\Delta_{v,c}\sigma$ on the last day. Signals are therefore correlated only through the shared σ and g_m, not through any physiological coupling. Stable patients are always in the stable mode; for the other scenarios the generator chooses the deteriorating mode with probability $0.4+0.5\sigma$ (capped at 0.95) and the recovering mode otherwise. Finally, it clips SpO_2 at 100%, steps and sleep at zero and HRV at 1 ms.

This construction matters for interpreting results: a classifier trained on it learns to invert these deltas, so held-out accuracy measures recovery of the generator's own labels, not clinical validity.

We rejected two candidate sources after inspection: a 5,000-row "HF clinical records" variant turned out to be a resampled copy of the same $\sim$300 patients (only 203 unique platelet values, 3,680 exact duplicate rows), and a public wearable dataset described as HF data contained general fitness data from 16 participants.

B. Scenario taxonomy

We fixed five deterioration scenarios before modeling (Table IV). Each maps to a specific set of Pulse actions, which we chose so that the simulated physiology, not merely the label, differs between scenarios.

C. Scenario model: classifier and severity regressor

Two Random Forests [44] (300 trees, default depth) share one 29-column feature matrix: five clinical features (age, sex, BMI, EF, NT-proBNP) and, for each of six wearable signals, the first-7-day mean, last-7-day mean, their difference and the 21-day linear slope (24 features). We drop the trend-mode column used during synthesis to prevent label leakage, and split patients 70/15/15 (stratified by scenario) into training, validation and test sets.

TABLE III
DATASETS AND THEIR ROLES

Source	**Role**	**Access**
MIMIC-IV [38]	Reference statistics (11,837 HF admissions); real-outcome test (separate extract: 17,129 HF ICU admissions)	PhysioNet credentialed [40]
Chicco & Jurman [34]	EF, creatinine, sodium statistics	Public, CC BY 4.0
NHANES	Height/weight/BMI by sex	Public survey
PerHeart [41]	Real home data, 27 HF patients aged 67–94; pipeline replay	Ethics-approved, Zenodo
Zigong HF cohort [42]	Real-outcome tests (2,001 admissions)	PhysioNet, DUA
Multi-pathology BCG [43]	Measured SV/CO for 2 patients	figshare

TABLE IV
SCENARIO TAXONOMY AND PULSE REPRESENTATION

Scenario	**Wearable signature**	**Pulse actions**
stable	Flat trends	EF-derived modifiers (continuous mode); none beyond the EF $\leq 40\%$ condition (fresh mode)
fluid_overload	Rising weight, rising HR	Venous compliance reduced with severity
cardiac_stress	HR, HRV changes under exertion; preserved EF	Modifiers + `Exercise`
deconditioning	Falling steps, mild HR drift	Resistance and compliance modifiers; no exertion
acute_ deterioration	Rapid multi-signal decline; reduced EF	Modifiers + HR multiplier + `Exercise`

D. Physiology engine and personalization

1) Circuit model: Pulse models the circulation as a lumped-parameter network, the hydraulic analogue of an electric circuit: nodes carry pressures, paths carry flows, vessel segments are resistors and elastic segments are capacitors, so that for each segment

$$Q = \frac{\Delta P}{R}, \qquad C = \frac{dV}{dP}, \tag{8}$$

with arterial segments represented as Windkessel elements (a compliance in parallel with a resistance) [45]. Pulse assembles these branch equations with Kirchhoff's laws and solves the network by modified nodal analysis at every time step [21]. At a compliant node, conservation of volume and (8) with constant C give

$$C\,\frac{dP}{dt} = \frac{dV}{dt} = \sum Q_{\text{in}} - \sum Q_{\text{out}}. \tag{9}$$

For a single two-element Windkessel fed by an inflow $Q(t)$ and draining through R to a downstream pressure taken as zero, (9) becomes

$$C\,\frac{dP}{dt} = Q(t) - \frac{P}{R}, \tag{10}$$

so that when inflow stops in diastole the pressure decays as $P(t) = P_0\,e^{-t/RC}$, the behavior Frank's model was introduced to explain [19].

2) Heart model: Each heart chamber is a time-varying compliance whose inverse, the elastance, follows a model adapted from Stergiopulos *et al.* [46]:

$$E_v(t) = (E_{\max,v} - E_{\min,v})\,\frac{f(t)}{f_{\max}} + E_{\min,v}, \tag{11}$$

$$f(t) = \frac{(t/\alpha_1 T)^{n_1}}{1 + (t/\alpha_1 T)^{n_1}} \cdot \frac{1}{1 + (t/\alpha_2 T)^{n_2}}, \tag{12}$$

where $E_{\max,v}$ and $E_{\min,v}$ are the maximum and minimum ventricular elastances, T is the cardiac-cycle duration and $\alpha_1, \alpha_2, n_1, n_2$ are shape parameters; chamber compliance is $1/E_v(t)$. This is the time-varying-elastance description of the ventricle [20] in circuit form. The cyclic change in compliance produces the pressure and volume changes that drive flow through the network, and Pulse computes stroke volume, cardiac output and arterial pressures from the circuit quantities at each time step. Ejection fraction is therefore an *output* of this model, not an input.

3) How the engine applies modifiers: We therefore act on the parameters of (8)–(12). In Pulse 4.3.1 the `CardiovascularMechanicsModification` multipliers and the `ChronicVentricularSystolicDysfunction` condition enter at the start of every cardiac cycle as [47]

$$E_{\max,L} \leftarrow \beta_E\,k_{\text{HF}}\,m_{\text{SV}}\,E^0_{\max,L}, \tag{13}$$

$$E_{\max,R} \leftarrow \beta_E\,m_{\text{SV}}\,E^0_{\max,R}, \tag{14}$$

$$f_{\text{HR}} \leftarrow m_{\text{HR}}\,f_{\text{HR}}, \qquad R_i \leftarrow m_R\,R_i, \tag{15}$$

$$C_i \leftarrow m_C\,C_i, \qquad C_j \leftarrow m_V\,C_j, \tag{16}$$

where $E^0_{\max}$ are the configured maximum elastances, β_E is the baroreflex elastance scale (1 with the reflex off), $k_{\text{HF}} = 0.27$ when the condition is present and 1 otherwise, f_{HR} is the heart-rate driver frequency (after baroreflex, chemoreflex and drug effects, and bounded by the patient's minimum and maximum heart rate), i runs over systemic resistance and compliance paths and j over venous compliance paths. The engine squares m_{SV} when it exceeds 1, which our mapping never does. The stroke-volume multiplier therefore acts on contractility (maximum elastance), not on stroke volume directly, and the condition is a fixed 73% cut in left-ventricular maximum elastance with no severity parameter. Fig. 2 shows where each lever acts on the circulation.

4) EF-to-engine mapping: We map EF and severity onto these multipliers and, for reduced EF, the binary `ChronicVentricularSystolicDysfunction` condition (the fixed elastance reduction k_{HF}). With d_{EF} from (3), the combined deficit is

$$x = \begin{cases} 0.5\,\sigma, & \text{EF} \le 40\%, \\ \max\left(d_{\text{EF}},\, 0.8\,\sigma\right), & \text{EF} > 40\%, \end{cases} \tag{17}$$

where the condition is applied ($k_{\text{HF}} = 0.27$) in the first case and not ($k_{\text{HF}} = 1$) in the second. The base multipliers are

$$\begin{aligned} m_{\text{SV}} &= \max(0.5,\, 1 - 0.5x), \\ m_R = m_C &= \max(0.6,\, 1 - 0.3x). \end{aligned} \tag{18}$$

When we apply the condition, the continuous multiplier represents only the *additional* acute severity (the first case of (17)), because stacking both produced cardiovascular collapse in validation. Since $x \le 1$, $m_{\text{SV}} \ge 0.5$ and $m_R = m_C \ge 0.7$; with the condition, $x \le 0.5$, so the left maximum elastance stays at or above $0.27 \times 0.75 \approx 0.20$ of its configured value before baroreflex scaling.

5) Scenario actions: In the default fresh mode, the four non-stable scenarios add their own multipliers to the same action; a multiplier named by both the scenario and (18) takes the scenario's value:

$$\begin{aligned} &\text{FO:} && m_V = \max(0.5,\, 1 - 0.4\sigma), \\ &\text{CS:} && m_{\text{HR}} = 1 + 0.3\sigma, \quad e = \sigma, \\ &\text{DC:} && m_R = m_C = \max(0.85,\, 1 - 0.1\sigma), \\ &\text{AD:} && m_V = \max(0.4,\, 1 - 0.5\sigma), \\ & && m_{\text{HR}} = 1 + 0.4\sigma, \quad e = 0.6\sigma, \end{aligned} \tag{19}$$

with the scenario abbreviations of Table II, where e is the intensity of an `Exercise` action, clipped to $[0.1,\, 0.5]$; Section IV shows that the intensity the engine tolerates under disease modifiers is patient-specific and can be lower. A fresh run starts from the patient file and condition, stabilizes for 60 s, applies the actions and advances 10 min.

6) Engine boundary: Pulse rejects patients outside age 18–65 and BMI 16–30. We clamp demographics only at the engine boundary, so the patient's true EF and severity still drive the simulation while the simulated body is a proxy:

$$a' = \text{clip}(a,\, 18,\, 65), \qquad w' = \text{clip}\left(w,\, 16.5\,h^2,\, 29.5\,h^2\right), \tag{20}$$

with height h (m) held fixed; the half-unit BMI margins avoid rejections from floating-point rounding.

E. Hemodynamic features and risk score

From the first and last samples of each run we extract HR rise, MAP drop, cardiac-output (CO) drop, a compensation flag (stroke volume held $\ge 95\%$) and an instability flag ($\text{MAP}_{\text{end}} < 65$ mmHg):

$$\Delta\text{HR} = \text{HR}_{\text{end}} - \text{HR}_{\text{start}}, \qquad \Delta\text{MAP} = \text{MAP}_{\text{start}} - \text{MAP}_{\text{end}},$$

$$\Delta\text{CO} = 100\,\frac{\text{CO}_{\text{start}} - \text{CO}_{\text{end}}}{\text{CO}_{\text{start}}},$$

$$F_{\text{comp}} = \mathbf{1}\left[\frac{\text{SV}_{\text{end}}}{\text{SV}_{\text{start}}} \ge 0.95\right], \qquad F_{\text{inst}} = \mathbf{1}[\text{MAP}_{\text{end}} < 65]\,. \tag{21}$$

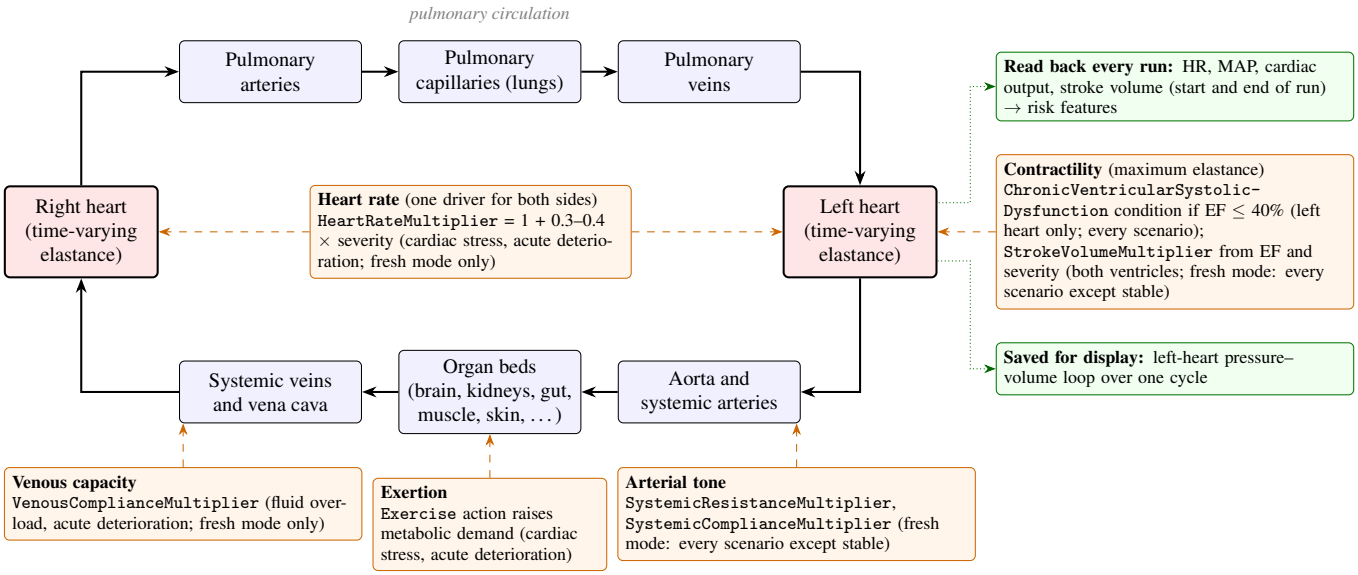


Fig. 2. Simplified Pulse circulation, showing where the pipeline acts (orange, dashed) and what it reads back (green, dotted). Scenario mappings are for fresh mode; the note at the bottom gives the continuous mode, which both scenario-test experiments used. Names and mappings are from `scenario_file.py` and `patient_file.py`.

In fresh mode and on a patient's first continuous day, the run includes stabilization, so these deltas span the healthy baseline to the end of the encounter; on later continuous days they span only that day's encounter, from the re-issued modifiers to its end (Section VII-F).

The primary risk score (Table V) normalizes each to $[0, 1]$ against a clinically anchored full scale and forms a weighted sum:

$$
\begin{aligned}
s_{\text{HR}} = \tfrac{1}{3} N_2(70 + \Delta\text{HR}), \qquad s_{\text{MAP}} &= \text{clip}\left(\frac{\Delta\text{MAP}}{27.5}, 0, 1\right), \\
s_{\text{CO}} &= \text{clip}\left(\frac{\Delta\text{CO}}{30}, 0, 1\right), \\
s_{\text{comp}} = 1 - F_{\text{comp}}, \qquad s_{\text{inst}} &= F_{\text{inst}}, \\
A = 0.15\, s_{\text{HR}} + 0.20\, s_{\text{MAP}} + 0.20\, s_{\text{CO}} & \\
+0.15\, s_{\text{comp}} + 0.30\, s_{\text{inst}}, &
\end{aligned}
\tag{22}
$$

where $A = \sum_i w_i s_i$ is the acute score, $N_2(h)$ is the NEWS2 heart-rate score [49] (3 for $h \le 40$, 1 for $40 < h \le 50$, 0 for $50 < h \le 90$, 1 for $90 < h \le 110$, 2 for $110 < h \le 130$ and 3 above 130 bpm), applied to an assumed resting HR of 70 bpm plus the rise, and $27.5 = 92.5 - 65$ mmHg spans the healthy anchor to the instability threshold.

During validation the score was structurally blind to fluid overload, whose danger appears as an already-congested baseline rather than an acute change, so we added a sixth input:

$$
\begin{aligned}
r &= \max\left(\textstyle\sum_i w_i\, s_i,\ s_{\text{baseline}}(\text{MAP}_{\text{start}})\right), \\
s_{\text{baseline}} &= \text{clip}\left(\frac{92.5 - \text{MAP}_{\text{start}}}{27.5}, 0, 1\right),
\end{aligned}
\tag{23}
$$

where s_i are the five acute sub-scores and s_{baseline} scores how far the simulated starting MAP lies below a healthy 92.5 mmHg anchor. We call a day's dominant mechanism *baseline* when $s_{\text{baseline}} > A$ and *acute* otherwise; the C3 guard uses this label (Algorithm 1). Buckets are LOW ($r < 0.35$), MODERATE ($r < 0.65$) and HIGH; these cut-points are an engineering choice, not a clinical citation.

Two consequences follow from (22) and (23). First, no two acute components can reach HIGH: the two largest weights sum to 0.50, so $A \ge 0.65$ requires at least three non-zero components. Second, the baseline term alone gives MODERATE when $\text{MAP}_{\text{start}} \le 82.9$ mmHg and HIGH when $\text{MAP}_{\text{start}} \le 74.6$ mmHg.

We later found that the fluid-overload scenario itself is misspecified (Section VII-F): it lowers venous compliance without adding circulating volume. The baseline-deficit term therefore compensates for a flawed scenario rather than reflecting correctly simulated congestion, and its improvement on the batch (30/30 LOW to 29/30 MODERATE) should be read in that light.

We keep an XGBoost regressor [50] on the same features as a secondary comparison only (5-fold CV MAE 0.089 ± 0.020, R^2 0.828 ± 0.091; $n = 117$ is too small to make it primary).

F. Pre-specified and post-hoc design decisions

We changed several scoring elements after seeing results. Table VI records when we fixed each one relative to the data we evaluated it on, so that results on tuning data are not mistaken for out-of-sample evidence.

TABLE V
RISK-SCORE COMPONENTS AND CLINICAL ANCHORS

Component	Weight	Anchor (citation)
HR rise	0.15	NEWS2 heart-rate bands [49]
MAP drop	0.20	Healthy baseline to MAP 65 mmHg [51]
CO drop	0.20	30% decline = full scale [52], [53]
Compensation failure	0.15	Frank–Starling failure (binary)
Instability flag	0.30	MAP < 65 mmHg [51]
Baseline deficit	max()	Starting MAP below 92.5 mmHg anchor

TABLE VI
WHEN EACH SCORING ELEMENT WAS FIXED

Element	Status	Independent evaluation
Risk weights and anchors	Pre-specified (literature)	All later data
Bucket cut-points 0.35/0.65	Pre-specified (engineering)	All later data; not clinically validated
Baseline-deficit term	Post-hoc on 117-run batch	Scenario test, PerHeart run 3, MIMIC-IV, Zigong
NYHA feature removed	Post-hoc on live batch	45 new live patients; PerHeart
C3 component guard	Post-hoc on dev seeds	Held-out seeds 45–47
WATCH tier (`decide_alert`)	Post-hoc	Replay on both seed sets only
Scenario expectations	Pre-specified (git)	—

G. Staging, deterioration rate and projection

NYHA staging is rule-based: AHA/ACC structural criteria (LVEF $\leq$ 40% or age-adjusted NT-proBNP above cut-off) [3] gate whether symptoms have a structural basis, and the simulated exertion response places the patient in classes I–IV.

The system computes a deterioration rate from per-vital slopes normalized to population standard deviations per day; projection re-runs the full Pulse pipeline at linearly extrapolated severities for 7, 14 and 30 days.

H. Service, alerts and continuous state

The FastAPI service exposes eleven endpoints over seven tables: patients, clinical reports, wearable readings, simulation runs and risk assessments, plus two tables for the continuous mode (saved engine states and state resets). The service persists all results, so every read endpoint is a database read and never blocks on the engine; in a timing test of 50 calls per endpoint, the five read endpoints answered in a median 2.7–4.2 ms. Table VII lists the technology stack.

Missing EF or NT-proBNP defaults to a documented healthy value (EF 62%, NT-proBNP 100 pg/mL), always flagged and never silently blended with measured values.

A continuous-state mode serializes the Pulse engine state after each day and resumes from it the next day, instead of rebuilding the patient from scratch (Section IV). Its first day applies only the EF- and severity-derived modifiers; each later day re-issues them and, for the two exertion labels (cardiac stress and acute deterioration), adds Pulse's `Exercise` action, which stays active in the saved state. The system applies the scenario-specific extras in Table IV (venous compliance, heart-rate multiplier, systemic resistance and compliance) only in the default fresh mode. Both scenario-test experiments (Section V) used the continuous mode.

Algorithm 1 Daily alert decision with the C3 guard (`decide_alert`)

Input: classifier severity $\hat{\sigma}$; engine status s; today's risk bucket b_t, dominant mechanism d_t and instability flag $F_{\text{inst},t}$; yesterday's streak k_{t-1} and flag ι_{t-1}.
Output: level $\in \{\text{ALERT}, \text{WATCH}, \text{NONE}\}$.

C3 bookkeeping, whenever a risk assessment is written:
if $b_t = \text{HIGH}$ **and** $d_t = \text{baseline}$ **then** $k_t \leftarrow k_{t-1} + 1$; $\iota_t \leftarrow \iota_{t-1}$
else $k_t \leftarrow 0$; $\iota_t \leftarrow$ false
if $F_{\text{inst},t} = 1$ **then** $\iota_t \leftarrow$ true
if $s =$ unstable **then return** ALERT **if** $\hat{\sigma} > 0.15$ **else** NONE *classifier fallback*
if $s =$ valid **then**
  if $b_t = \text{HIGH}$ **then**
    if $k_t > 3$ **and not** ι_t **then return** WATCH *C3: downgrade*
    return ALERT
  if $b_t = \text{MODERATE}$ **then return** WATCH
  return NONE
return NONE *engine not yet run*

Algorithm 2 One day of the continuous-state twin (`run_daily_continuous_pipeline`)

Input: patient's wearable readings and clinical reports; saved state S_{t-1} (none on the first day or after a reset), with the EF used for it.
Output: saved state S_t; risk assessment and alert for day t.

$W \leftarrow$ last 21 days of wearable readings; **if** fewer than 21 **then** stop
if a clinical report arrived after S_{t-1} **then** take EF and NT-proBNP from it
else carry forward the EF stored with S_{t-1} *EF changes only with a new report*
  missing EF/NT-proBNP $\rightarrow$ 62%, 100 pg/mL, flagged
$(\hat{c}, \hat{\sigma}) \leftarrow$ classifier and regressor on features of W and clinical values
$M \leftarrow$ `CardiovascularMechanicsModification`$(m_{\text{SV}}, m_R, m_C)$ from (17)–(18)
if S_{t-1} is none **then**
  build the clamped patient file (20); add the condition **if** EF $\leq 40\%$
  stabilize 60 s; apply M; advance 600 s
else
  load S_{t-1} (`EngineStateFile`); re-issue M *state does not carry it*
  if $\hat{c} \in$ {cardiac stress, acute deterioration} **then** apply `Exercise` at $\text{clip}(f_{\hat{c}}\,\hat{\sigma}, 0.1, 0.5)$, $f = 1.0$ or 0.6
  advance 600 s
if the engine failed **then** record a failed run; keep S_{t-1}; alert by line 5 of Algorithm 1; stop
$S_t \leftarrow$ `SerializeState`; store EF, $\hat{\sigma}$, simulation time and report id with S_t
compute (21)–(23) on today's output; run Algorithm 1

Algorithm 1 gives the alert logic and Algorithm 2 one day of the continuous-state twin. An engine run counts as *unstable* if it failed, or if it was an acute-deterioration run with severity in the documented crash zone $[0.6, 0.85]$, even when it completed; for unstable runs the alert falls back to the classifier's severity. On resume the engine cannot add or remove the `ChronicVentricularSystolicDysfunction` condition, so a new EF that crosses 40% is reported as a mismatch until the twin is reset.

Five research features (continuous mode, BCG-derived compliance, a measured heart-rate baseline, alert hysteresis and scenario-label persistence) sit behind flags that are off by

TABLE VII
TECHNOLOGY STACK

Layer	Component
Physiology engine	Kitware Pulse 4.3.1 (`kitware/pulse:4.3.1` image, amd64) [13]
Machine learning	scikit-learn 1.9.0 Random Forest [44], [54]; XGBoost 3.2.0 (secondary) [50]
Back end	Python 3.11, FastAPI, SQLAlchemy, Pydantic; SQLite (Postgres-portable)
Front end	React 19 + Vite; hand-built SVG charts
Deployment	Docker Compose (Postgres, back end, front end)
Testing	pytest (353 test cases from 334 test functions, run on every pull request), Playwright browser checks, GitHub Actions CI



Fig. 3. Dashboard pressure–volume loop for the patient in Fig. 4 (EF 30%): left-heart pressure against volume over one steady-state cardiac cycle at the end of the day's run. Synthetic demonstration patient; not a real person.

default; with all flags off, a golden-file test checks that every API response is identical to the version without them.

The React dashboard shows, for each patient, the simulated pressure–volume loop (Fig. 3), the alert level beside the classifier's own signal (Fig. 4) and the risk-score history (Fig. 5).

IV. Engineering Challenges and Solutions

Using Pulse as an HF twin surfaced problems that neither the documentation nor offline evaluation revealed. Table VIII lists those specific to the engine and the modeling; general software and infrastructure issues are in Appendix Table XXII.

A. *Pulse instability at high severity*

In a 150-patient batch (30 per scenario), 117 runs completed (Fig. 6). All 33 failures occurred in the two scenarios that add an `Exercise` action; 30 were engine crashes and 3 were timeouts, and failed patients had high severity (mean 0.75 and 0.72), so missingness is *not at random* with respect to severity.

Engine logs for three independent crashes showed an identical chain: fatigue, hypoxia and tachycardia, then renal hypoperfusion and cardiovascular collapse, ending in a fatal negative heart-chamber volume and an irreversible-state event at 154.1–154.3 s.

We applied single-variable interventions to one crash case that repeated identically on rerun (Table IX). Gaps and ramps only delayed the crash; either stressor alone completed. The

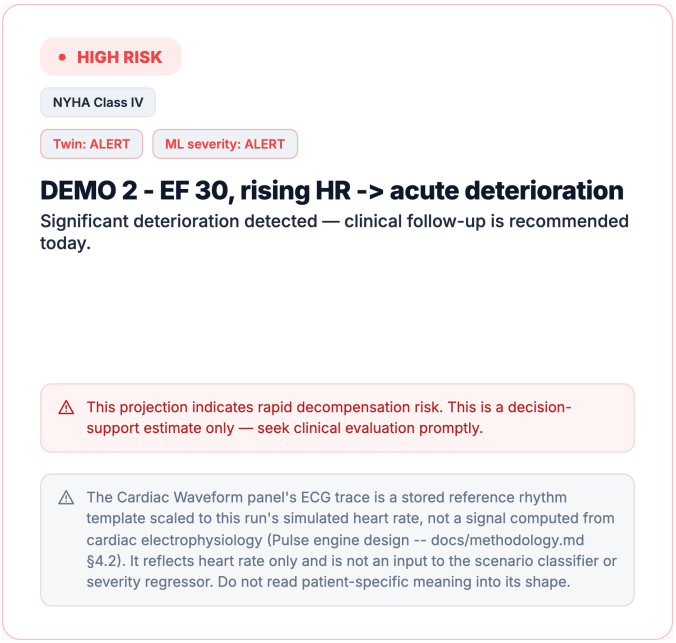


Fig. 4. Dashboard alert view: risk bucket, NYHA class and the twin's `decide_alert()` level (ALERT, WATCH or NONE) beside the classifier-severity signal, with the ECG-template caveat. Here both signals fire; when exactly one fires, a note says which. Synthetic demonstration patient; not a real person.

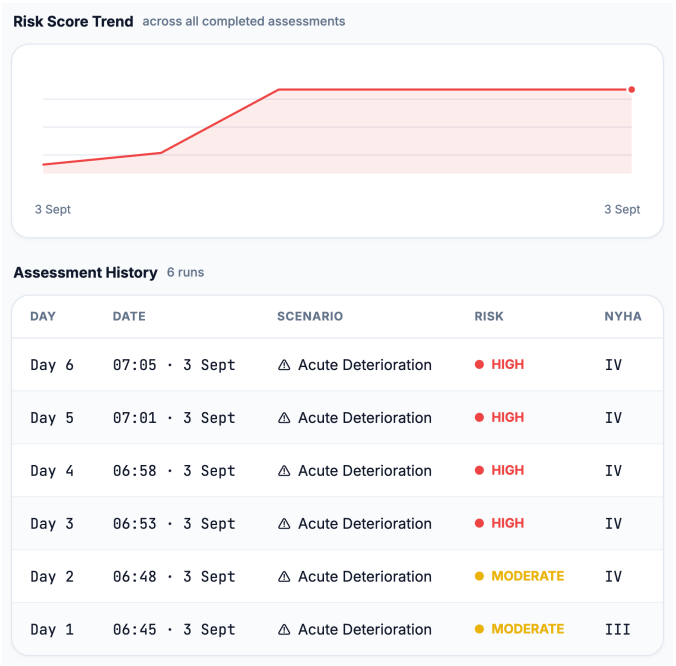


Fig. 5. Dashboard risk timeline for a patient in continuous mode over six daily runs, recorded on 3 September 2026 with earlier code: MODERATE on days 1–2 and HIGH on days 3–6, with the predicted scenario acute deterioration on every day. The dashboard chart has no y-axis values. Synthetic demonstration patient; not a real person.

failure is therefore a sustained combined-load boundary whose safe exercise intensity varies by patient (about 0.22 for one patient and between 0.125 and 0.25 for another), so no single

TABLE VIII
ENGINE AND MODELING PROBLEMS, ROOT CAUSES AND SOLUTIONS

Problem	Root cause (how found)	Solution	Evidence
Pulse rejected scenario files	Actions/conditions need `PatientAction/AnyCondition` wrappers; found from protobuf parse errors	Builders emit the wrapped schema	All 5 scenarios run
Runs exited 1 with time mismatch	`CardiovascularMechanicsModification` without `Incremental:true` silently re-stabilizes, consuming extra time	Set `Incremental:true`	Clean exit, exact end time
Collapse at low EF (negative chamber volume)	Binary EF condition stacked with a second EF-derived cut	Continuous multiplier carries only the additional acute severity	Collapse no longer occurs
Real patients outside Pulse range	Engine hard-rejects age > 65, BMI > 30; real HF cohorts are older	Clamp demographics only at the engine boundary; flag per patient	All 16 replayed PerHeart patients needed the age proxy
Deconditioning looked like cardiac stress	Both used `Exercise`; HR 150–164 in both	Removed exertion from deconditioning	Distinct HR/MAP profiles (Table XIII)
Fluid overload always scored LOW	Risk only measured change within a run; fluid overload is a shifted baseline	Added baseline-deficit term with `max()`	30/30 LOW $\rightarrow$ 29/30 MODERATE; scenario later found mis-specified
Live severity error 0.271 vs. 0.048 offline (NYHA-era model)	NYHA ordinal used in training but unknown at inference, defaulted to class I	Removed NYHA from features; retrained	Live MAE 0.271 $\rightarrow$ 0.026
Engine crashes at high severity	Disease modifiers + sustained exertion exceed engine's sustainable load (Table IX)	Characterized; failures recorded, never dropped	Same failure chain in 3 logged crashes (154.1–154.3 s)
Resumed state drifted	Disease modifiers are not carried by serialized state	Re-issue modifiers on every resume	Drift 6.5% $\rightarrow$ 0.000%
HIGH risk persisted after exertion days	Pulse resets the baroreflex MAP set point after 7 min off target and stores it in the patient record, which the saved state carries (saved-state comparison, engine source, replay)	Documented, not yet fixed; restoring the set point alone saturates the reflex	Set point 95.4 $\rightarrow$ 65.1 mmHg; HR $\approx$46 bpm on HIGH days
Engine segfault in continuous mode	Pulse Python SDK `initialize_engine()` crash (gdb trace); earlier "thread conflict" diagnosis disproved	Switched to the CLI scenario driver	Day 1$\rightarrow$3 runs clean
Resumed runs flagged as crashes	Driver's end-time check ignores elapsed state time; schema field renamed (`Mode`, not `Type`)	Resume-aware crash filter; corrected schema	Exact match vs. continuous
SpO_2 output always 0.0	Persists with Pulse's bundled patient and a request byte-identical to a known-good one; likely arm64 emulation	Never used as a feature	Excluded
ECG looked computed	Pulse ECG is a stored template scaled by HR	Labeled display-only in UI/API	Caveat in every response

TABLE IX
ISOLATION EXPERIMENTS ON ONE CRASH CASE (SEVERITY 0.731)

Intervention	Outcome
Control (exact reproduction)	Crash at 154.32 s
60 s gap before exertion	Crash at 212.02 s
120 s gap before exertion	Crash at 272.92 s
Exertion ramped in 4 steps	Crash at 301.14 s
Exertion alone, no disease modifiers	Completes (660 s)
Disease modifiers alone, no exertion	Completes (660 s)
Exertion intensity halved	Completes (660 s)

safe constant exists.

The crash case's patient body was not recorded. With the code current at submission, the same severity, modifiers and exertion completed all 660 s on two other patient bodies, so we cannot tell whether the crash depends on that body or on a later change in the engine path.

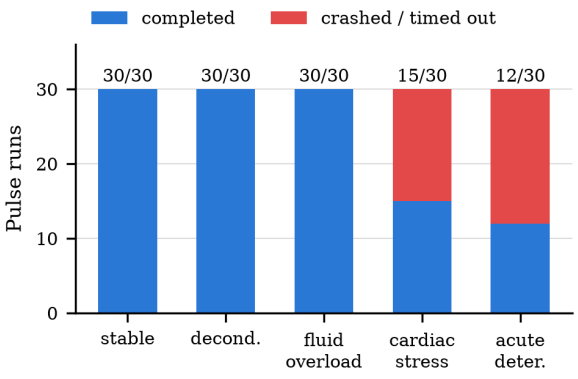


Fig. 6. Pulse batch completion by scenario (150 runs). All failures occur in the two scenarios that combine disease modifiers with an `Exercise` action.

B. *Two failure mechanisms with opposite character*

Failures were not a single phenomenon. Mechanism 1 (engine crash) depended on the patient's own severity and

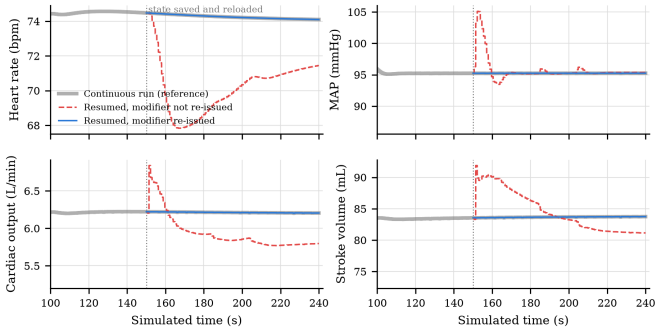


Fig. 7. State resume, rerun on 12 October 2026 through the production command-line path (`cli_state_runner.resume_and_advance`; EF 56.4, severity 0.46). The state was saved at 150 s and reloaded. Without re-issuing the disease modifier, HR, CO and SV drift away from the uninterrupted run; with it re-issued, the resumed run lies on the reference. Final values differ slightly from Table X (e.g., continuous CO 6.205 vs. 6.145 L/min), which came from an earlier run.

repeated identically when we reran the same case, so retrying an identical scenario did not help. Mechanism 2 (timeouts and connection-pool exhaustion) depended on how many other patients were running: at 9 workers, 5/11 patients hit connection timeouts and 6/11 hit the 180 s engine timeout; at 4 workers 8/11 still timed out; at 2 workers there were no failures. A bare completion rate conflates the two; we report the mechanism with each rate.

C. Continuous state: carrying a patient forward

Rebuilding a patient every day discards the physiological state the twin is meant to track. Pulse can serialize its full state (about 2.33 MB, saved in 0.31–0.44 s and loaded in 0.43 s) and resume it, and with an `Exercise` action the resumed run's final heart rate, MAP, cardiac output and stroke volume matched an uninterrupted run to every printed digit.

With our pipeline's disease modifier, they did not match (Table X): the serialized state does not carry the modifier. Re-issuing it on every resume restored the match, to 15 significant figures in all four outputs (we did not compare the serialized states themselves), which also requires storing the last-used EF and severity alongside the state. A rerun through the command-line driver used in production shows the same pattern over time (Fig. 7).

A later end-to-end test crashed inside the Pulse Python SDK; isolating it showed a native crash in the SDK's engine initialization even with no machine-learning code loaded, disproving an earlier thread-conflict diagnosis, so we rebuilt the feature on the command-line driver.

D. Other modeling corrections

Three further corrections followed from running the real system rather than offline checks.

1) Severity (classifier output) and risk score (simulation-derived) share a $[0, 1]$ range but not a scale: for acute deterioration the risk score's minimum (0.491) exceeded the severity mean (0.385). The projection module had applied a risk-score calibration to severity; we separated the two calibrations, and the system now reports the two quantities side by side, never fused.
2) The classifier's label occasionally flipped between days for genuinely ambiguous trajectories; a categorical persistence rule (window chosen by sweep) now stabilizes the reported scenario.
3) We tried adding rolling-window curvature features ("Option A"). They kept accuracy unchanged against a production model retrained in the same environment (90.7% both) but worsened severity MAE from 0.0473 to 0.0506 on held-out data, so we reverted them.

TABLE X
STATE RESUME FIDELITY (PATIENT EF 56.4, SEVERITY 0.46; FINAL VALUES AT 240 S)

Run	HR	MAP	CO	SV
Continuous (reference)	74.13	95.22	6.145	82.89
Resumed, modifier not re-issued	71.51	95.34	5.743	80.32
difference	−3.5%	+0.1%	−6.5%	−3.1%
Resumed, modifier re-issued	74.13	95.22	6.145	82.89
difference	0.000%	0.000%	0.000%	0.000%
Two continuous runs (control)	identical			

V. RESULTS

We evaluate in layers of increasing realism. This section reports the layers that use synthetic patients; Section VI reports the tests against real data. Fig. 8 shows the design, and Table XI summarizes what each layer tests and what it does not. We report the following scenarios, in this order:

- *Offline model test* (Section V-A): 300 held-out synthetic patients.
- *Engine scenario check* (Section V-B): one simulated patient per scenario at severity ≈0.5, and the 117 completed batch runs.
- *Live API validation* (Section V-C): 25, then 50 attempted synthetic patients replayed day by day through the live API.
- *Pre-specified 21-day scenario test, Experiment 1* (Section V-D): ten scripted stories on development seeds 42–44 and held-out seeds 45–47; on the same runs, an ablation of the alert definition (Section V-E) and an analysis of false alerts and label errors (Section V-F).
- *Experiment 2* (Section V-G): the same stories with ejection fraction 58–67% on new seeds 48–53, followed by the mechanism behind high-risk days in both experiments (Section V-H).

The metrics are:

- *Accuracy* and *severity MAE*: the share of held-out patients assigned the correct scenario, and the mean absolute error of predicted severity on its 0–1 scale.
- *Label accuracy*: the share of scenario-test patient-days on which the predicted scenario matches the story's intended scenario.

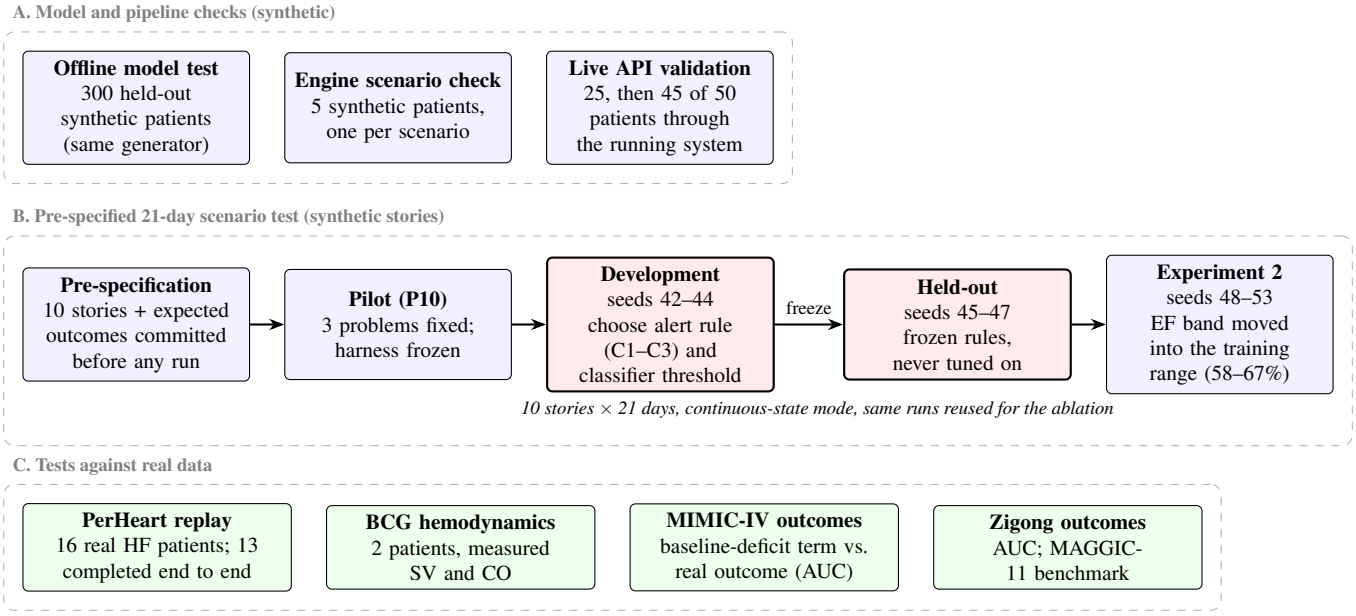


Fig. 8. Evaluation design. (A) Model and pipeline checks on synthetic patients. (B) The pre-specified 21-day scenario test: outcomes committed before any run, a pilot on P10, rule selection on development seeds 42–44, the frozen rules on held-out seeds 45–47, and Experiment 2 on seeds 48–53. (C) Tests against real data.

TABLE XI
EVALUATION LAYERS AND THEIR SCOPE

Layer	Data	Main result	Does not test
Offline model test	300 held-out synthetic patients	91.3% accuracy, severity MAE 0.047	Real patients
Engine scenario check	5 synthetic patients in Pulse	Physiologically distinct scenario responses	Clinical accuracy
Live API validation	25 then 45 synthetic patients via live API	25/25 completed; live MAE 0.026 after fix	Real patients
Pre-specified scenario test	10 scripted stories × 6 seeds × 21 days	21/30 deteriorating series caught (3/5 stories in all seeds); label accuracy 17.6%	Real deterioration events
Ablation (same runs)	30 deteriorating, 18 stable series	C3 guard cut false alerts 61–68%; tuned classifier matches or beats the Pulse-based alert	Clinician interpretability
Experiment 2 (EF in range)	Same stories, EF 58–67%, 6 new seeds	Label accuracy 37.1%; 24/30 caught, 4.8 FA/100 pd	Fluid-overload and acute labels
Real-patient replay	PerHeart, 16 HF patients	13/16 completed end to end	Outcomes; patients' own physiology (age proxy)
BCG hemodynamics	2 patients, measured SV/CO	Simulated CO 45–46% below measured	Cohort-level accuracy; trends
Real-outcome tests	MIMIC-IV, Zigong	Baseline-deficit term: AUC 0.596 on MIMIC-IV (age alone 0.591); weak on Zigong (0.548)	Day-by-day early warning; the full twin

- *Sensitivity*: deteriorating patient-seed series (five stories × six seeds = 30) with at least one alert on or after the story's onset day; *strict detection* also requires the first alert to fall on or after onset.
- *Pre-onset alerts per 100 days*: alert days before onset in the deteriorating series, per 100 such days.
- *False alerts per 100 patient-days* (FA/100 pd): alert days in the three stay-quiet stories (18 series), per 100 of their patient-days; *specificity* counts stay-quiet series with no alert at all.
- *AUC*: area under the ROC curve for real outcomes, with bootstrap 95% CIs (Section VI-C).

A. Offline model accuracy

On the 300-patient held-out test set, the classifier reached 91.3% accuracy (274/300, bootstrap 95% CI 88.0–94.3%; macro-F1 0.91) and the regressor severity MAE 0.047 (RMSE 0.061). Per-class precision and recall were 0.88–0.95, with no class collapsed into another. The earlier 92.3% figure belonged to a version that included NYHA class, which we removed for the reason in Table VIII.

Because test patients come from the same generator, these figures measure how well the model inverts the generator, not how it performs on independent trajectories; the pre-specified cohort below is the more relevant estimate (17.6% day-by-day label accuracy with the cohort's EF band below the training range, 37.1% with it inside).

TABLE XII
SCENARIO MODEL ON 300 HELD-OUT SYNTHETIC PATIENTS

Variant	Accuracy	Severity MAE	RMSE
With NYHA feature (superseded)	92.3%	0.048	0.063
Production (clinical + wearable)*	**91.3%**	**0.047**	**0.061**
+ curvature features (reverted)	90.7%	0.051	0.065
Clinical features only	50.3%	0.171	0.219

*Frozen model used for every reported result. The other rows come from earlier training runs; retraining the production configuration in a different software environment gave 90.7%, so differences of about one point are within run-to-run variation.

TABLE XIII
SIMULATED RESPONSE BY SCENARIO (SEVERITY ≈0.5)

Scenario	HR (bpm)	MAP (mmHg)	Note
stable	71→70	95→95	Flat
deconditioning	71→74	95→95	Mild drift
fluid_overload	72→72	77→77	Congested baseline; CO +13%
cardiac_stress	71→164	95→67	Healthy compensation (EF 67%)
acute_deter.	72→132	78→52	SV cannot rise (EF 26.6%)

TABLE XIV
PRE-SPECIFIED SCENARIO COHORT

ID	Story	Group
P01	Stable patient, ordinary life	stay quiet
P02	Salty weekend (weight bump, recovers)	edge case
P03	Slow, quiet weight gain	edge case
P04	Fluid overload building (sev. 0.20→0.50)	catch
P05	Gradual deconditioning (sev. 0.20→0.35)	catch
P06	Cardiac stress, exertion episodes	catch
P07	Sudden deterioration, days 10–13	catch
P08	Stressful fortnight, healthy heart	stay quiet
P09	Active, stable patient	stay quiet
P10	Everything goes wrong (sev. 0.20→0.45)	catch

Cardiac stress (preserved EF) and acute deterioration (reduced EF), the pair hardest to separate from simulated HR/MAP alone, were cleanly separated because EF is an input. Removing all 24 wearable features cost 40 points of accuracy (Table XII), confirming that the wearable trend, not the clinical snapshot, carries most of the signal.

B. Engine-level scenario check

We simulated each scenario for one patient at severity ≈0.5 (Table XIII). Responses were physiologically distinct. The clearest contrast is between cardiac stress and acute deterioration: both raised heart rate, but the healthy heart (EF 67%) raised cardiac output from 5,910 to 9,948 mL/min with stroke volume maintained, while the failing heart (EF 26.6%) could not increase stroke volume (63→67 mL) and MAP fell to 52 mmHg. We did not tune this contrast; it emerged from the EF-driven modifiers.

Across the 117 completed batch runs, outputs also followed the severity given to the engine where each scenario's mechanism predicts (Spearman ρ): heart rate (0.92 and 0.89) and cardiac output (0.81 and 0.92) rose with severity in cardiac stress and acute deterioration, and MAP fell with it in cardiac stress (-0.73); stable runs showed no dose response, as designed, and fluid-overload MAP barely moved ($\rho = 0.15$), consistent with that scenario's missing volume-loading mechanism (Section VII-F).

The engine's left-heart pressure–volume signal was also physiologically plausible (volume 62.2–143.0 mL, pressure 5.8–136.8 mmHg at EF 45%), and the dashboard shows it as a PV loop. Fig. 9 shows loops recorded during the scenario test for the same simulated patient before and after deterioration: by day 14 the loop has shifted to larger volumes (end-diastolic volume from about 150 to 220 mL) with a lower peak pressure (about 139 to 121 mmHg). The ECG beside it is a stored template that Pulse scales with heart rate; it does not respond to EF or scenario and is displayed for reference only.

C. Live API validation

We replayed twenty-five synthetic patients (five per scenario) day by day through the live API with no mocking: 25/25 completed and all 25 were classified correctly, but severity MAE was 0.271, far from that model's offline 0.048 (the version that still used NYHA class; the production model's offline MAE is 0.047). Per-patient inspection showed predictions compressed into a narrow 0.02–0.19 band, which we traced to the NYHA train/inference skew.

After we removed the feature and retrained, live revalidation reached severity MAE 0.008 on five patients and, expanded to 50 attempted (45 completed), MAE 0.0264 (95% bootstrap CI 0.0194–0.0352) with 100% scenario accuracy. The five failures were all high-severity exertion scenarios, matching Mechanism 1.

D. Pre-specified 21-day scenario test (Experiment 1)

We wrote ten patient stories and committed their expected outcomes to the project's git repository before any simulation (commit `38c3520`, 3 October 2026, 02:04 IST; the first pilot run followed about 50 minutes later). This is pre-specification under version control, not registration in a public registry.

The cohort (Table XIV) has five stories that should be caught, three that should stay quiet and two edge cases. We ran each story for 21 days in continuous-state mode on development seeds 42–44 and, after selecting a rule, on held-out seeds 45–47. Fig. 10 shows the wearable inputs for five of the stories.

A pilot run exposed a design confound (EF matched to story severity) and two harness bugs (day-to-day noise scaled to the current value instead of the population SD, and per-process salting of Python's `hash()` making seeds non-reproducible); we fixed all three and froze the harness before the main runs, recording every amendment.

1) Two alert definitions: Our first analysis alerted on the classifier's severity (> 0.15) and caught all five deteriorating patients, but also alerted on all three stable ones (58.7 false

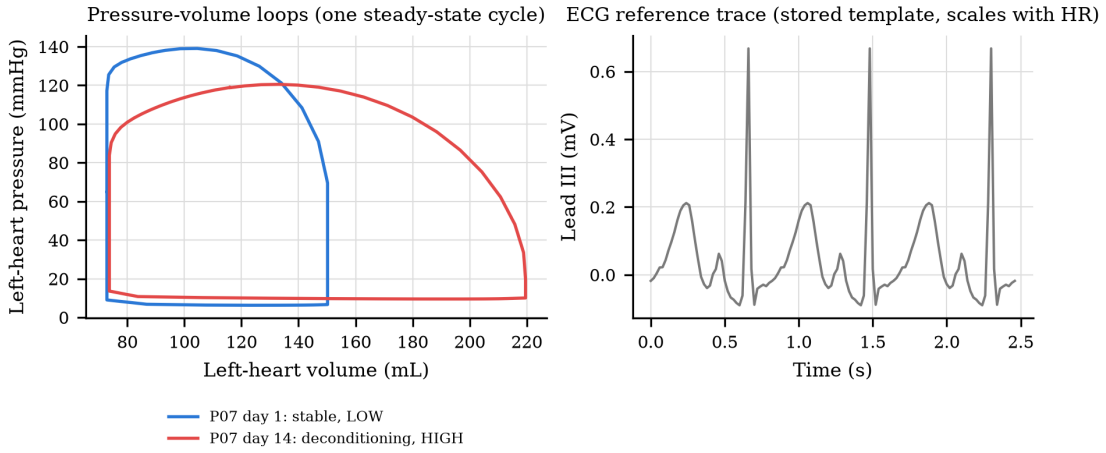
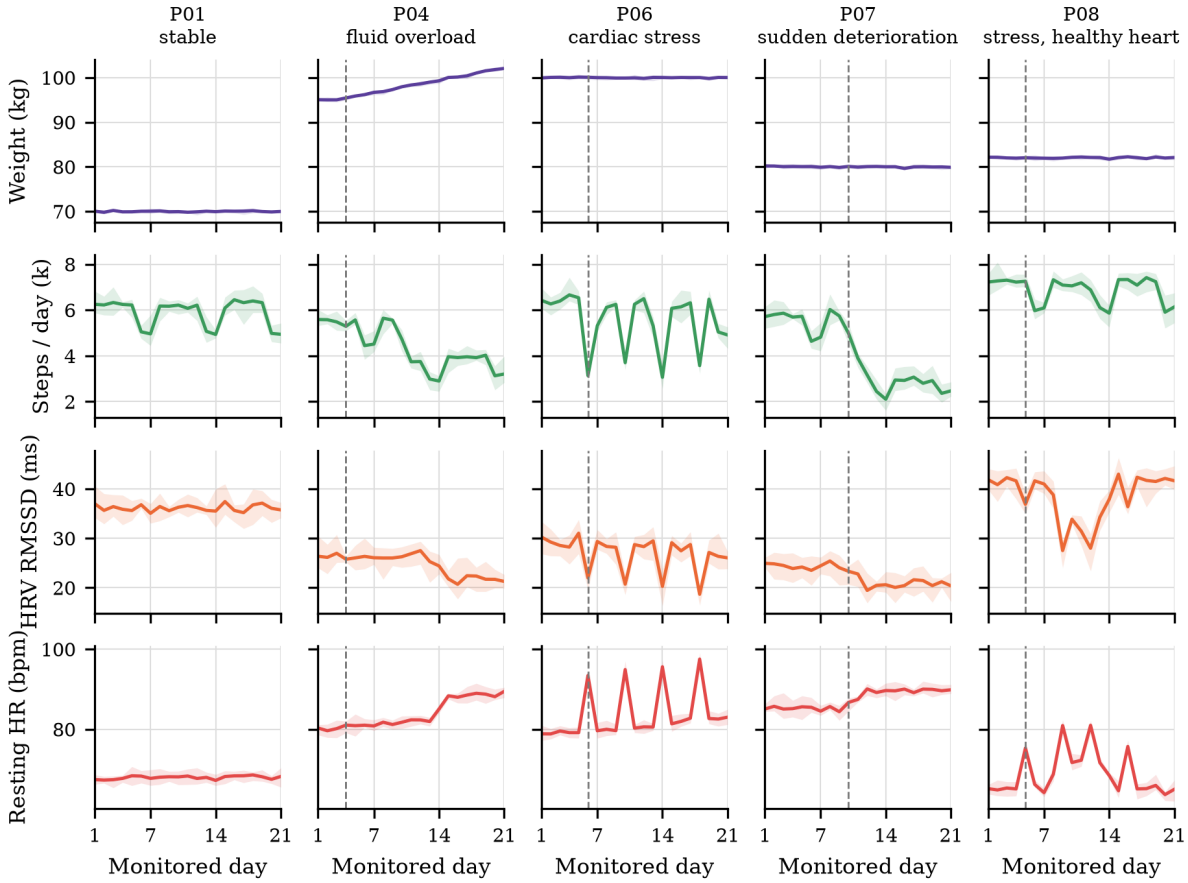


Fig. 9. Pulse output saved during the scenario test (P07, seed 45). Left: left-heart pressure–volume loop over one steady-state cardiac cycle on day 1 (LOW risk) and day 14 (HIGH risk). Right: the engine's ECG reference trace, a stored waveform scaled by heart rate rather than a computed electrophysiological signal.



Mean and range over six seeds (42-47). Dashed line: designed onset of the story's change.

Fig. 10. Wearable inputs the classifier receives for five of the ten stories (weight, steps, HRV and resting HR; mean and range over seeds 42–47). Dashed line: the story's designed onset.

alerts per 100 patient-days). A code audit then showed that the dashboard did not use this signal at all: it displayed

TABLE XV
ALERT RULES: DETECTIONS AND FALSE ALERTS

	Dev. (42–44)		Held-out (45–47)	
Rule	**Caught**	**FA/100**	**Caught**	**FA/100**
Baseline (risk bucket HIGH)	3/5	17.99	3/5	30.16
C1: HIGH for 3 consecutive days	3/5	13.76	3/5	25.40
C2: add sustained-MODERATE watch	3/5	17.99	3/5	30.16
C3: component guard	**3/5**	**5.82**	**3/5**	**11.64**

the risk-score bucket (HIGH). When we re-analysed under the dashboard's definition, the system detected three of five deteriorating patients (P06, P07, P10) with 18.0 false alerts per 100 patient-days.

P05 (deconditioning) produced near-zero values on every risk component on all 63 patient-days, and P04 (fluid overload) did not congest in some seeds, so neither had any signal for an alert rule to act on.

2) Alert rule selection: We specified three candidate rules before computing any of them (Table XV). C3 downgrades a sustained HIGH that is driven only by the baseline-deficit term with no corroborating instability, the diagnosed cause of P08's persistent false alarm. It kept all three detections and cut false alerts by 67.6% on development seeds and 61.4% on held-out seeds, where we never tuned it.

3) Unified alert decision: We then merged the two alert signals into a single `decide_alert()` function, now the only source of alert state in the API and dashboard. It has three levels: ALERT (risk bucket HIGH and not downgraded by C3, or a crashed/failed engine run via the classifier-severity fallback); WATCH (risk bucket MODERATE, or a HIGH streak that C3 downgrades); and NONE. C3 therefore downgrades but never silences: a HIGH risk bucket can never map to NONE.

Replaying the saved scenario-test runs through the deployed functions reproduced the detection pattern on both seed sets (P06, P07 and P10 at ALERT in every seed; P01 and P09 quiet in every seed). P08's false alarm, previously ALERT for 8–14 consecutive days, fell to about 3 ALERT days per seed, with the remaining elevated days shown as WATCH. For P04 held-out seed 47, which never reached HIGH, 20 of 21 days now read WATCH, making a previously invisible fluid-overload trajectory visible for the first time.

E. Ablation: does the physiology layer improve alerting?

The premise of the twin is that simulating physiology adds something the classifier alone does not. We tested this on the same saved runs by changing only the alert definition (Table XVI, Fig. 11). At its default threshold (severity above the stable range, 0.15) the classifier detected all 30 deteriorating series but alerted in every stable series, and passing its output through Pulse and the C3 guard reduced false alerts from 60.1 to 8.7 per 100 patient-days at a sensitivity of 70%. Requiring the first alert to follow onset lowers this to 14/30: P07, whose onset is on day 10, alerted before onset in all six seeds.

TABLE XVI
ALERT RULES ON THE SAME RUNS (ALL SIX SEEDS)

Alert rule	**Sensitivity**	**FA / 100 pd**
Weight rule	2/30 (0.02–0.21)	0.0
Classifier, severity > 0.15	30/30 (0.89–1.00)	60.1 (44.4–73.8)
Classifier, severity $> 0.55^a$	30/30 (0.89–1.00)	8.5 (4.0–13.8)
Classifier, severity $> 0.60^a$	28/30 (0.78–0.98)	2.4 (0.0–5.6)
Label only (exertion scenario)	19/30 (0.46–0.78)	18.8 (7.7–31.0)
Pulse risk bucket HIGH	21/30 (0.52–0.83)	24.1 (10.6–38.6)
Deployed (C3 guard)	21/30 (0.52–0.83)	8.7 (3.4–16.1)
Deployed, WATCH or ALERT	22/30 (0.56–0.86)	29.1 (14.6–46.3)

Wilson 95% CIs for sensitivity; bootstrap 95% CIs (2,000 resamples over series) for false alerts per 100 patient-days in the stable stories. [a]Threshold chosen with knowledge of these data; see Table XVII for a development/held-out comparison.

TABLE XVII
CLASSIFIER THRESHOLD SELECTED ON DEVELOPMENT SEEDS, TESTED ON HELD-OUT SEEDS

Held-out seeds 45–47	**Sensitivity**	**FA / 100 pd**
Classifier, severity > 0.55 (dev-selected)	15/15 (0.80–1.00)	11.6 (4.2–20.1)
Deployed Pulse system (C3, dev-selected)	10/15 (0.42–0.85)	11.6 (3.2–25.4)

That comparison, however, favors the physiology layer, because we never tuned the classifier's threshold. Sweeping the classifier's severity threshold traces a curve that lies above and to the left of every Pulse-based rule (Fig. 11): at 0.55 it detected 30/30 series with 8.5 false alerts per 100 patient-days, and at 0.60 it detected 28/30 with 2.4.

A rule using only the predicted label (alert on an exertion scenario) performed similarly to the Pulse risk bucket (19/30 series, 18.8 false alerts), consistent with the label, more than the simulated physiology, determining when Pulse reports high risk.

To avoid tuning on the test data, we selected the classifier threshold on development seeds alone, using a rule fixed in advance (the lowest threshold whose development false-alert rate does not exceed the deployed system's, 5.8), exactly as we selected C3, and evaluated it once on held-out seeds (Table XVII). The rule chose 0.55. On held-out seeds the classifier detected all 15 deteriorating series and the deployed Pulse system 10, at the same false-alert rate (11.6 per 100 patient-days). In this case study, therefore, the physiology layer did not improve alerting over a properly thresholded classifier.

A conventional weight rule ($\geq$ 2 kg within 3 days) never false-alerted but detected only 2/30 series, because the scripted fluid build-up (about 0.4 kg per day) never reaches 2 kg within three days.

1) Absolute burden and sample size: Even the best rules carry a clinically heavy absolute rate: 8.7 false alerts per 100 patient-days means roughly one false alert every 11 or 12 days for each stable patient, and one every 9 days on the held-out seeds alone.

With ten stories and six seeds, the confidence intervals are wide (Table XVI); we present these results as a controlled case

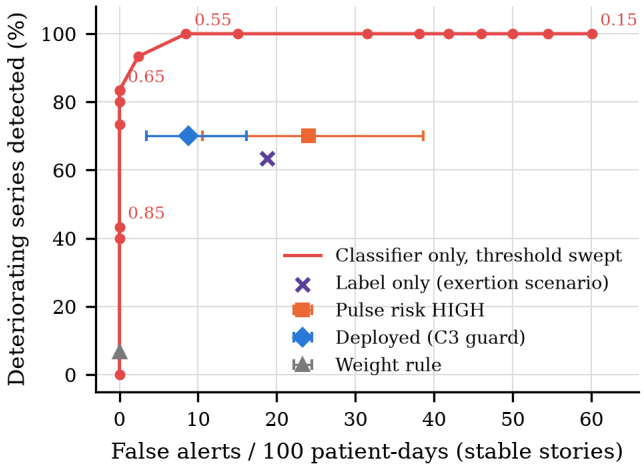


Fig. 11. Sensitivity versus false-alert burden on the same saved runs. The red curve sweeps the classifier's severity threshold (labels mark thresholds); points show the Pulse-based rules (95% CIs), a label-only rule and a weight rule. The tuned classifier dominates the Pulse-based rules on this cohort.

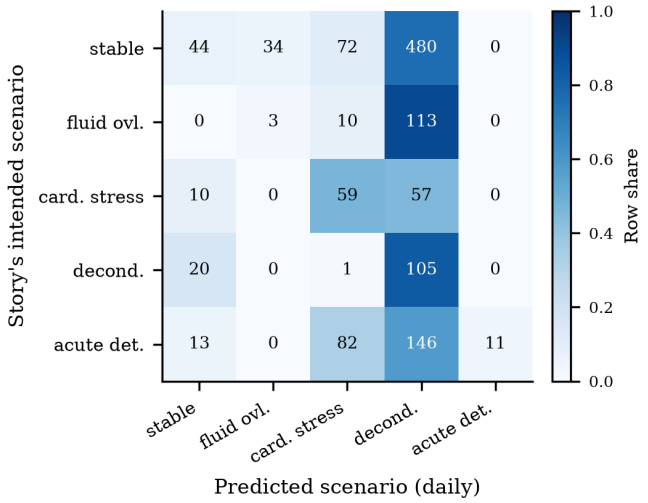


Fig. 12. Daily scenario predictions against each story's intended scenario, all 1,260 patient-days (counts; colour is row share). Deconditioning dominates every row.

study, not as estimates of performance in a patient population. Engine failures did not distort them: only 3 of 1,260 patient-days failed, and every metric was identical whether we scored failed days as alerts (the deployed fallback) or as no alert.

F. Why false alerts occur

Feature importances show the severity regressor relies on resting-HR delta and slope (63.2% combined) and step-count trends (27.2%), the signals that ordinary day-to-day variation moves most, while EF ranks tenth (0.6%). Adding Gaussian noise at one population SD to a stable patient's window moved predicted severity from 0.11 to 0.43, crossing the stable-range threshold (Table XVIII).

The classifier mostly predicted the wrong scenario. Day-by-day scenario-label accuracy on this cohort was 17.6% (222/1,260 patient-days; 19.8% if days before a story's onset are scored as "stable"), below the 20% expected from guessing among five classes and far below the 91.3% in-distribution test accuracy. The confusion matrix (Fig. 12) shows why: "deconditioning" was predicted on 71.5% of all days, including 480 of 630 days of stable stories and 113 of 126 days of the fluid-overload story; acute deterioration was recognized on only 11 days.

A likely cause is distribution shift: the cohort's neutral EF band (48–57%) sits below the EF range of the training "stable" class (51.3–70.7%) and near the deconditioning mean; Experiment 2 tests this explanation directly.

In the continuous mode used here, the label changes the simulation only through exertion (Section III): stable, fluid-overload and deconditioning labels produce the same simulation at a given EF and severity. The label errors therefore mattered where they added an exertion label to a story without one or withheld it from a story that needed it. The risk score still reached HIGH on 28% of deconditioning-labeled days (255/898) and 83% of cardiac-stress-labeled days (185/224);

TABLE XVIII
SENSITIVITY OF PREDICTED SEVERITY TO ADDED WEARABLE NOISE

Patient	0×SD	1×SD	1.5×SD	2×SD
P01 (stable)	0.115	0.428	0.544	0.560
P10 (deteriorating)	0.777	0.570	0.622	0.850

once exertion days have occurred, the carried-forward engine state keeps the score high on later days even when the label reverts to deconditioning (Fig. 13; quantified in Section V-H). The ablation results therefore describe the pipeline as it behaves with these label errors, not a physiology layer driven by correct scenarios.

G. Experiment 2: the same stories with an in-distribution EF band

To test the distribution-shift explanation, we repeated the scenario test with three changes and nothing else, recorded with predictions for every outcome before any run (commit `f353bd0`): the cohort's EF band moved from 48–57% to 58–67%, inside the training "stable" range; P07's onset moved from day 10 to day 4, like the other stories; and one unscored warm-up day was added so that every scored day, including day 1, is a full simulation day. We left the models, risk score, thresholds, C3 guard and feature flags unchanged and used six new seeds (48–53). The pre-specified plan also disclosed a classifier-only dry run on development seeds and fixed sensitivity and pre-onset alert rate as co-primary outcomes.

All 60 series completed (3 of 1,260 patient-days failed, as in Experiment 1). Table XIX compares both experiments. Label accuracy rose from 17.6% to 37.1% (468/1,260; 41.5% with pre-onset days scored as "stable"), and deconditioning fell from 71.5% to 52.9% of days.

The deployed alert caught 24/30 deteriorating series (21/30 before), alerted on 10.0 per 100 pre-onset days (31.7 before),

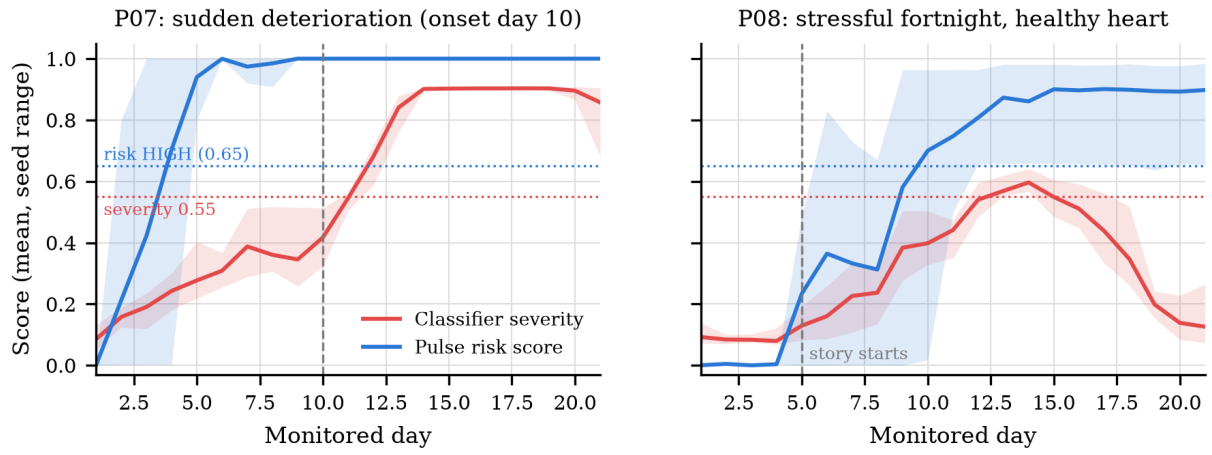


Fig. 13. Daily classifier severity and Pulse risk score (mean and range over six seeds) for a deteriorating story (P07) and a stable story with a stressful fortnight (P08). In P07 the risk score saturates days before the designed onset; in P08 it stays high after the stress episode ends while classifier severity recovers.

and its false alerts fell from 8.7 to 4.8 per 100 patient-days, with P01 and P09 now silent in every seed.

The results met seven of the ten outcome predictions. Three were not met: strict detection rose to 17/30 instead of falling, because the number of series alerting before onset stayed at 7/30 rather than rising as predicted; P05 (deconditioning) was detected in 2 of 6 series against a predicted 0–1, because those two series received an exertion label (on day 1 and day 5), the mechanism in Section V-H; and the classifier's false alerts at 0.55 were 10.1 per 100 patient-days, just above the predicted ceiling of 10.

The classifier still never labelled fluid overload (P04) or acute deterioration (P07) correctly: in the training data those classes occur only at EF $\leq$48%, so at the new EF band acute deterioration was predicted on only 8 days (all P10) and fluid overload on none.

The main finding is unchanged: the classifier at its frozen threshold of 0.55 still detected all 30 series, each with its first alert on or after onset, while the Pulse-based alert, now with fewer false alerts, missed six.

H. How the twin's high-risk days arise

The saved runs of both experiments show one mechanism, replicated on new seeds and a new EF band; we found it in a descriptive analysis after both experiments. No HIGH-risk day preceded a series' first exertion label (0 of 462 HIGH days in Experiment 1, 0 of 475 in Experiment 2). Series with an exertion label reached HIGH in 30 of 32 and 30 of 31 cases, series without one in 0 of 28 and 0 of 29, and the first HIGH day came 0–1 days after the first exertion label in 23 of 30 series in each experiment.

After the first exertion label, the engine settles into a low-pressure, slow-heart state: later days without an exertion label started at a median MAP of 66 mmHg, against 95 mmHg before any exertion label (Fig. 14); HIGH days ended at a median heart rate of 46 bpm, with 78% (Experiment 1) and 75% (Experiment 2) below 50 bpm; and the baseline-deficit term, which scores starting MAP, decided 452 of 462 and 471 of 475 HIGH days (Fig. 15).

TABLE XIX
EXPERIMENT 1 VS. EXPERIMENT 2 (SAME STORIES, EF 58–67%, SEEDS 48–53)

Outcome	Exp. 1	Exp. 2	Predicted
Label accuracy	17.6%	37.1%	30–45%
Label accuracy, pre-onset as stable	19.8%	41.5%	35–50%
Sensitivity[a]	21/30	24/30	18–26
Pre-onset alerts / 100 days (series)[a]	31.7 (7/30)	10.0 (7/30)	10–30 (6–14)
Strict detection[b]	14/30	17/30	6–12*
P05 series detected	1/6	2/6	0–1*
Stable series with no alert	10/18	12/18	11–13
FA / 100 pd, deployed	8.7	4.8 (1.6–7.9)	<8.7
Classifier > 0.55, strict	30/30	30/30	≥26
Classifier > 0.55, FA / 100 pd	8.5	10.1 (4.8–15.6)	≤10*

Deployed alert (C3 guard) unless stated; failed days scored as no alert (as alerts: sensitivity 25/30, strict 18/30). [a]Co-primary. [b]First alert on or after onset. *Prediction missed. Bootstrap 95% CIs over series.

We traced this state to Pulse's baroreflex in an exploratory analysis after both experiments. Pulse 4.3.1 resets the reflex's MAP set point whenever MAP stays more than 5% from it for 7 minutes, moving it 35% of the way toward the current MAP, and it stores the set point in the patient record [48]. Under the disease modifiers, simulated exertion holds MAP well below the set point, so the set point falls with each reset. Because the continuous mode saves the patient record with the engine state, the lowered set point carries into every later day. In one series (P08, Experiment 2, seed 48), the set point fell from 95.4 to 65.1 mmHg over three exertion days. Replaying the first of those days from the saved day-8 state reproduced the fall in four resets over 30 simulated minutes (95.4 to 65.6 mmHg). With the set point near 65 mmHg, the reflex holds MAP there

and slows the heart.

The saved states show the same pattern across series. In Experiment 2 and on the held-out seeds of Experiment 1 (the runs whose states we kept), 99% and 98% of HIGH days began with a set point below 85 mmHg (median 64 mmHg), against 2% and 8% of other days (median 95 mmHg).

Neither ending the exercise nor restoring the set point undid the state. From the saved day-13 state of the same series, deleting the `Exercise` action left MAP at 67 mmHg after 20 minutes, while heart rate fell from 46 to 37 bpm. Restoring the set point to 95 mmHg made it worse. MAP now sat about 30% below the set point, beyond the 2.6% at which Pulse's reflex saturates. In saturation, unless blood volume is falling, Pulse moves heart rate away from its sympathetic target, so heart rate fell to 20 bpm and MAP to 53 mmHg within 20 minutes, while the set point reset back down to 76 mmHg. In the continuous mode, HIGH risk therefore reports an engine state that exertion triggers and the saved patient record preserves, not a gradual change in the simulated circulation.

This accounts for the label-only rule's resemblance to the Pulse risk bucket and for the ablation result, and it points to specific changes: keeping the baroreflex set point from carrying over between simulated days, and adding physiological inputs that do not pass through the label. Restoring the set point alone is not enough, as the saturation result shows.

VI. Validation

We tested the pipeline, or the components that can run without a 21-day wearable window, against three kinds of real data: home measurements (Section VI-A), measured hemodynamics (Section VI-B) and recorded outcomes (Section VI-C).

A. Real-patient replay (PerHeart)

The PerHeart pilot dataset [41] contains one month of real home measurements (pulse oximeter, scale, blood-pressure cuff) from 27 HF patients aged 67–94. Sixteen had at least 21 real overlapping days of heart rate, SpO_2 and weight; we fabricated no day, and unmeasured fields (sleep, HRV, steps, EF, NT-proBNP) took documented defaults with explicit flags. Every patient required the engine's age proxy.

Before the severity fix, all 16 completed but severities were compressed (0.078–0.148) and no patient reached HIGH, independently reproducing the NYHA skew on real data. After the fix, severities spread (0.093–0.516) and risk buckets were 4 LOW, 4 MODERATE and 5 HIGH (previously 4 LOW, 12 MODERATE, 0 HIGH), while three patients now crashed with the same engine failure signature.

The system staged all 16 as NYHA I, an artifact of the identical healthy EF/BNP defaults rather than a clinical finding. The one fluid-overload patient scored LOW because the EF default makes Pulse simulate a structurally normal heart, so the baseline-deficit term cannot fire; the API now names this mechanism in its caveat.

We searched the literature for a BNP-to-EF proxy and found none: the clinical relationship runs from EF to BNP, not the reverse.

TABLE XX
Simulated vs. Measured Hemodynamics (Ratio Sim/Real)

Subject	HR	Stroke volume	Cardiac output
14	0.94	0.58	0.54
102	0.63	0.87	0.55

B. Measured hemodynamics (BCG, two patients)

For two patients from a ballistocardiography dataset with echocardiographic volumes [43], we used BCG features to adjust the twin's vascular compliance and compared simulated stroke volume and cardiac output with measured values (Table XX). Simulated cardiac output was 45–46% below measured in both; 71.3% of patient 14's stroke-volume shortfall was already present from anthropometrics alone, before EF or BCG data entered the model.

For a synthetic 21-day deterioration anchored to these patients' real HR and EF, the classifier eventually assigned the correct label for both, but labeled patient 14 as fluid overload for days 6–10 before converging.

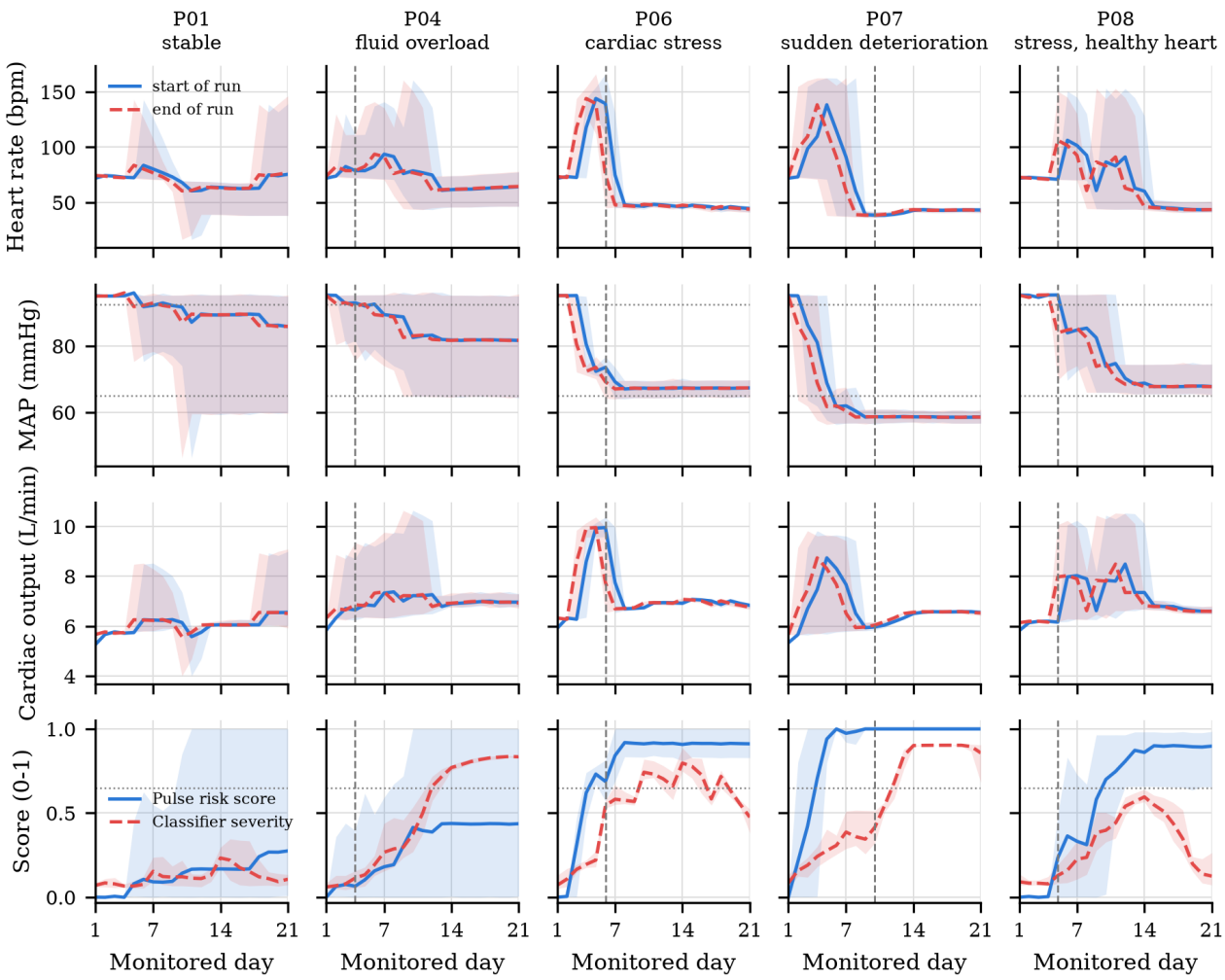
C. Real-outcome tests

Two real cohorts lack the 21-day wearable window and EF needed to run the whole twin, so we tested only components with a non-fabricated input path (Fig. 16, Table XXI). The baseline-deficit term is a function of starting MAP alone; applied to measured first-24-hour MAP in 17,129 MIMIC-IV HF ICU admissions, it predicted in-hospital mortality with AUC 0.596 (95% CI 0.585–0.608), with mortality rising from 8.9% to 21.2% across score deciles; age alone gave 0.591 (0.579–0.602) on the same admissions. This shows that low admission MAP carries modest prognostic information in an ICU population, about as much as age; it is not a test of the twin, which we did not run on these patients.

On the Zigong cohort [42] (6-month death or readmission), the same term reached 0.548 (0.525–0.573) on all 2,001 cleaned admissions but 0.533 (0.490–0.575) on the 625 complete cases, and a clinical-only variant of the scenario model 0.517 (0.473–0.562): weak signal at best.

As external context, an 11-of-13-variable MAGGIC score (point table checked against the published integer-score chart [30]) reached 0.604 on this cohort, and a model using laboratory panels reached 0.667, signal carried by renal and coagulation tests that a wearable-based system cannot obtain.

On the synthetic batch, the risk score agreed moderately with MAGGIC computed on the same patients (Spearman $r = 0.479$; tier agreement 38.5% against $\approx$33% by chance), indicating directional consistency rather than validity. Here we held MAGGIC's unmeasured inputs at fixed values, which shifts every patient's score equally and leaves the rank correlation unchanged; for heart-failure duration, whose chart label ("first diagnosis in the past 18 months") conflicts with the fitted model ("duration >18 months"), we followed the fitted model, on which the 38.5% tier agreement depends.



Each point is one day's Pulse run; mean and range over six seeds (42-47).
MAP dotted lines: 92.5 mmHg (healthy anchor) and 65 mmHg (instability threshold). Score dotted line: HIGH (0.65).
Grey dashed line: designed onset.

Fig. 14. Pulse output per day for five stories in Experiment 1 (mean and range over seeds 42–47): heart rate, MAP and cardiac output at the start and end of each day's run, and the Pulse risk score against classifier severity. Dotted lines: MAP 92.5 and 65 mmHg; HIGH cut-point 0.65. Dashed line: designed onset. In P06, P07 and P08 the start-of-run MAP falls to about 58–68 mmHg after the exertion days and stays there, which drives the baseline-deficit term.

VII. Discussion and Limitations

A. What the twin does well

The pipeline runs end to end, day after day, on synthetic and real home data, and continuous state makes it a persistent model of a patient, whose resumed outputs match an uninterrupted run, rather than a fresh simulation each day.

Its scenarios produce physiologically distinct responses through mechanism rather than labels (Table XIII), and in the exertion scenarios its outputs follow severity.

Because every alert decomposes into simulated hemodynamic components, we could diagnose and guard against a specific false-alarm pattern: the C3 guard cut false alerts by 61–68% without losing a detection, and it transferred unchanged to held-out seeds.

The evaluation also shows that we understand the system's behavior well enough to predict it: Experiment 2 met seven of the ten outcome predictions we committed before its runs, and the exertion mechanism replicated on new seeds and a new EF band.

B. Is the mechanistic layer circular?

For detection, largely, in its current form. The classifier's label selects the Pulse scenario and its severity scales it, and in the continuous mode every HIGH-risk day followed an exertion label, so the simulated physiology re-expresses the classifier's decision rather than observing new evidence. The ablation agrees: a label-only rule behaves like the Pulse risk bucket, and a classifier threshold tuned on development seeds

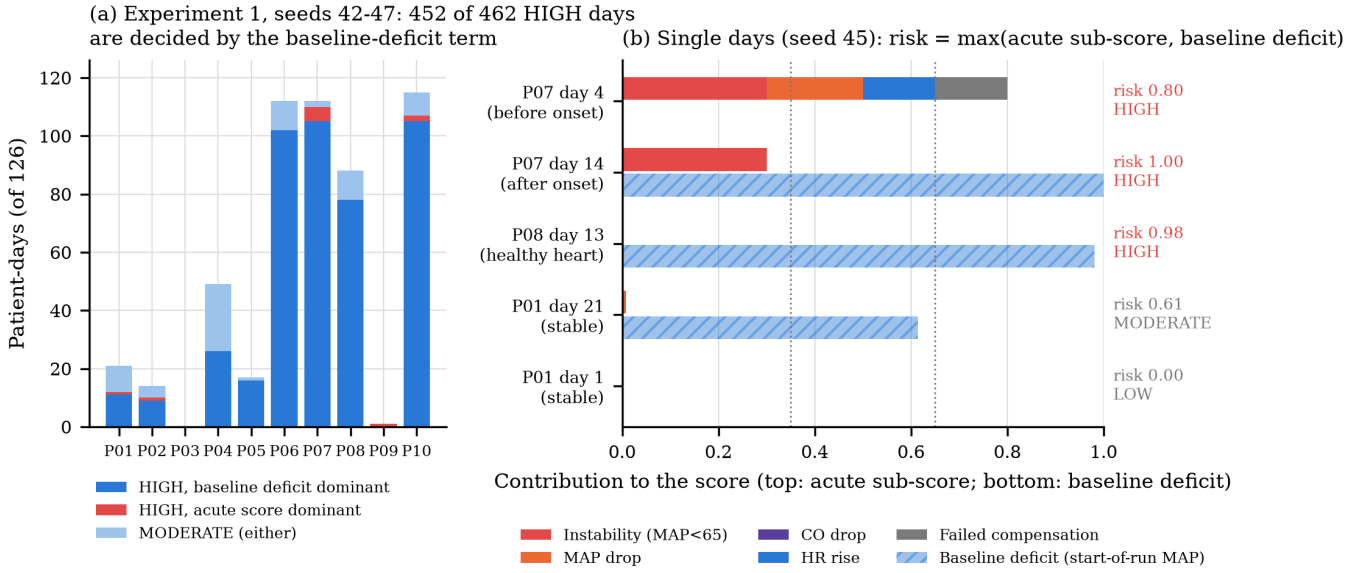


Fig. 15. (a) Risk bucket by story in Experiment 1 (seeds 42–47; 126 patient-days per story): the baseline-deficit term decided 452 of the 462 HIGH days. (b) Score components on single days (seed 45), where $r = \max(A, s_{\text{baseline}})$; dotted lines mark the 0.35 and 0.65 cut-points. P07 day 4, six days before the story's day-10 onset, is one of the 10 HIGH days decided by the acute score (a pre-onset alert); P08 day 13 (a healthy heart) is HIGH from the baseline deficit alone, the pattern the C3 guard targets.

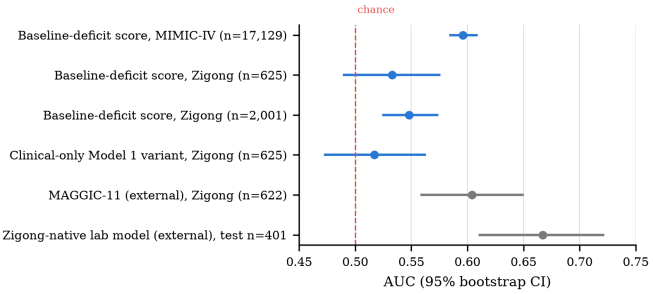


Fig. 16. Real-outcome discrimination. Blue: components of the twin. Grey: external benchmarks on the same cohort.

TABLE XXI
REAL-OUTCOME TESTS (AUC, 95% BOOTSTRAP CI)

Score / cohort	n	**AUC**
Baseline deficit, MIMIC-IV, in-hospital death	17,129	0.596 (0.585–0.608)
Age alone (reference), MIMIC-IV	17,129	0.591 (0.579–0.602)
Baseline deficit, Zigong, complete-case	625	0.533 (0.490–0.575)
Baseline deficit, Zigong, all cleaned	2,001	0.548 (0.525–0.573)
Clinical-only scenario model, Zigong	625	0.517 (0.473–0.562)
MAGGIC-11 (external)[a] [30], Zigong	622	0.604 (0.559–0.649)
Zigong lab model (external), test set	401	0.667 (0.611–0.721)

Zigong outcome: death or readmission within 6 months (41.3% event rate). Bootstrap: 2,000 resamples. [a]Excludes current smoking and heart-failure duration, which the Zigong data do not record; age taken as the midpoint of 10-year bands; point table checked against the published chart.

matched the deployed system's held-out false-alert rate while detecting more deteriorating series.

What the engine adds is different in kind: a hemodynamic account of each alert (pressure–volume loops, cardiac output, compensation), the component structure that made the C3 guard possible, and persistent state. We have not yet tested the value of these explanations to clinicians.

Two factors limit what the engine could add to detection here. First, the label was wrong on most days in Experiment 1; moving EF into the training range (Experiment 2) roughly doubled label accuracy and reduced false alerts, but did not close the gap to the tuned classifier. Second, the scenario that should carry congestion (fluid overload) is mis-specified.

Showing a detection benefit would require correct exertion labels, a continuous mode that does not carry Pulse's baroreflex set-point resets from day to day, a corrected fluid-overload scenario, and ideally physiological inputs (e.g., measured weight-driven volume changes) that do not pass through the classifier.

C. How personal is the twin?

Less than the term usually implies. On real patients, every PerHeart body was an age proxy, ejection fraction and NT-proBNP were defaults, simulated cardiac output was 45–46% below measured in the two BCG patients, and the severity regressor relies on resting-HR and step trends for about 90% of its importance (EF: 0.6%). The current system is best described as a population-parameterized physiological model driven by an individual's wearable trends; per-patient calibration of baseline hemodynamics is the main step toward a personal twin.

D. What it does not yet do

Five findings bound the claims. First, two of five deterioration types are invisible to the current alert because their

signatures never reach the simulated hemodynamics: deconditioning by design (no acute event), and fluid overload because the scenario reduces venous compliance without adding circulating volume, which mobilizes pooled blood rather than overloading the circulation. Second, the models are sensitive to ordinary wearable noise and to clinical inputs outside their training distribution. Third, on real outcomes, the only component testable without a wearable window (the baseline-deficit term) carries modest signal, comparable to age alone on MIMIC-IV and weak on Zigong; the full twin has not been tested against real events. Fourth, the physiology layer did not improve alerting over a tuned classifier. Fifth, even the best configurations produce roughly one false alert every 9–12 days per stable patient, a burden that would be hard to sustain clinically.

Our results therefore support the feasibility of running such a twin end to end, not an alerting benefit or clinical effectiveness.

E. Lessons for building physiology-engine twins

We found every major fault in Table VIII by running the real system, not by offline evaluation: the NYHA skew only appeared in live inference, the CORS fault only in a browser, the resume drift only with the production modifier, and the alert mismatch only by auditing what the dashboard actually displayed. Committing expectations before each run and holding out seeds guarded against post-hoc rationalization, and a pre-specified replication turned a diagnosis (distribution shift in EF) into a tested prediction. Reporting failure mechanisms rather than bare completion rates prevented a biased reading of the results.

F. Limitations

- Training data are synthetic. Only the population statistics trace to real aggregate data; the healthy EF profile, wearable baselines and scenario changes are assumptions. The pre-specified cohort is small (ten stories, six seeds).
- The fluid-overload scenario contains a modeling error rather than a mere limitation: it lowers venous compliance without adding circulating volume, which mobilizes pooled blood and moves the simulated hemodynamics in the improving direction as severity rises. Correcting it requires a volume-loading action, re-verification of engine stability, re-running the scenario validation, retraining the severity model and repeating the 21-day test; we have not yet done so.
- We tuned several scoring elements post hoc (Table VI).
- The continuous mode simplifies each resumed day to the EF- and severity-derived modifiers plus exertion, without the scenario-specific extras; from its second day the risk features measure that day's change, whereas we designed the score on fresh runs (in both experiments the baseline term, which reads starting MAP rather than a within-day change, decided 452 of 462 and 471 of 475 HIGH days, so the within-day features set few HIGH days; we did not count how often they set MODERATE days); and in a demonstration check, two of five patients' risk rose over three resumed days with an unchanged label, a drift we have not checked against the set-point resets in Section V-H.
- Pulse resets its baroreflex set point within minutes and stores it in the patient record, so a mode that chains short simulations over days inherits every reset (Section V-H). We have not yet found a way to manage the set point; restoring it alone saturates the reflex.
- Pulse's native range (age 18–65, BMI 16–30) excludes most HF patients, so we simulate older patients with a proxy body.
- Real EF and NT-proBNP were unavailable in the replay cohort, and their defaults mask the fluid-overload mechanism.
- The engine's SpO_2 output was unusable and its ECG is a template.
- Pulse is unstable when disease modifiers combine with sustained exertion, which under-represents high-severity exertion scenarios in all datasets.
- We do not model medication effects: Pulse includes a validated diuretic (furosemide), making diuretic modeling feasible, but has no beta-blocker or ACE-inhibitor/ARB substances.
- Risk-bucket cut-points and the deterioration-rate calibration constants are engineering choices, not clinically validated thresholds.
- The API has no authentication and is suitable only for research use.

VIII. Conclusion and Future Work

We built and evaluated HF-PULSE, a digital twin for HF home monitoring that interprets daily wearable trends through a mechanistic engine, scores the simulated hemodynamics with an interpretable risk score and, in its continuous mode, carries the engine's state from day to day. It runs end to end on synthetic and real home data, and we release it with frozen models, automated tests and a fresh-clone reproduction.

A pre-specified evaluation with held-out seeds and a replication showed what the physiology layer contributes and what it does not. Its component structure enabled an alert guard that cut false alerts by 61–68%, including on unseen seeds; correcting the cohort's EF band doubled label accuracy and nearly halved false alerts, as predicted; and every high-risk day followed an exertion label. On held-out seeds, however, the classifier with a properly selected threshold detected all deteriorating series at the same false-alert rate at which the Pulse system detected two thirds, so the engine's present value is explanation rather than detection. The reasons are specific (label errors outside the training distribution, an engine baroreflex set point that exertion lowers and the saved state carries forward, a mis-specified fluid-overload scenario, and dependence of the simulated scenario on the label) and therefore actionable.

Future work will:

TABLE XXII
SOFTWARE AND INFRASTRUCTURE ISSUES

Problem	Root cause (how found)	Solution	Evidence
Timeouts at 180 s	Some runs need ≈180 s under arm64 emulation	Documented margin; 2-worker cap	Re-attempts at 180.3–180.4 s
Failures under concurrency	SQLAlchemy pool (5+10) exhausted; CPU contention	Concurrency $9 \rightarrow 4 \rightarrow 2$	0 failures at 2 workers
Corrupted Pulse image	Local containerd-snapshotter extraction bug	Disabled snapshotter, re-pulled	Engine verified
Python 3.9 container errors	PEP 604 unions evaluated at class definition by SQLAlchemy/Pydantic	`Optional[...]` in model/schema files	Imports succeed
Dashboard could not call API	Browser enforces CORS (invisible to curl/TestClient)	CORS middleware (local only)	Found in browser test
Model drift across machines	Unpinned scikit-learn/numpy versions	Exact dependency pins; model SHA-256 checked before each run	Zero version warnings

1) fix the remaining scenario labels, through training data that cover fluid overload and acute deterioration at preserved EF or physiological inputs that bypass the label, keep Pulse's baroreflex set-point resets from carrying over between days, and re-run the ablation;
2) add a volume-loading mechanism to the fluid-overload scenario and a diuretic action using Pulse's furosemide model;
3) systematically map the engine's stability boundary across severity, scenario and body profile;
4) make the models robust to wearable noise and out-of-distribution clinical inputs;
5) personalize the twin's baseline heart rate and hemodynamics from measured data; and
6) evaluate day-by-day early warning on real longitudinal data with recorded HF events, the test that ultimately decides clinical value.

DATA AND CODE AVAILABILITY

Code (MIT license), synthetic data and derived results are in the project repository, https://github.com/Mrunmayi019/M2K-HF-PULSE. We do not redistribute raw MIMIC-IV, Zigong and PerHeart files, in line with their access agreements; PerHeart-derived outputs are shared under CC BY-NC-SA. We check the frozen models' SHA-256 hashes before every scenario-test run, and the repository holds both experiments' pre-specified expectations and per-day results. A fresh clone on a second machine reproduced the synthetic datasets, the model hashes and all five demonstration patients' risk, bucket, NYHA class and alert exactly.

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APPENDIX A
SOFTWARE AND INFRASTRUCTURE ISSUES

Table XXII lists issues that were general to the software stack rather than to the engine or the model; we report them for reproducibility.

REFERENCES

[1] S. L. James *et al.*, "Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017," *Lancet*, vol. 392, pp. 1789–1858, 2018, doi: 10.1016/S0140-6736(18)32279-7.

[2] G. Savarese *et al.*, "Global burden of heart failure: A comprehensive and updated review of epidemiology," *Cardiovasc. Res.*, vol. 118, no. 17, pp. 3272–3287, 2022, doi: 10.1093/cvr/cvac013.

[3] P. A. Heidenreich *et al.*, "2022 AHA/ACC/HFSA guideline for the management of heart failure," *Circulation*, vol. 145, 2022, doi: 10.1161/CIR.0000000000001063.

[4] S. C. Inglis *et al.*, "Structured telephone support or telemonitoring programs for patients with chronic heart failure," *Cochrane Database Syst. Rev.*, 2008, doi: 10.1002/14651858.CD007228.

[5] W. T. Abraham *et al.*, "Wireless pulmonary artery haemodynamic monitoring in chronic heart failure: A randomised controlled trial," *Lancet*, vol. 377, pp. 658–666, 2011, doi: 10.1016/S0140-6736(11)60101-3.

[6] J. Stehlik *et al.*, "Continuous wearable monitoring analytics predict heart failure hospitalization: The LINK-HF multicenter study," *Circ. Heart Fail.*, vol. 13, Art. no. e006513, 2020, doi: 10.1161/CIRCHEARTFAILURE.119.006513.

[7] N. Gautam *et al.*, "Artificial intelligence, wearables and remote monitoring for heart failure: Current and future applications," *Diagnostics*, vol. 12, Art. no. 2964, 2022, doi: 10.3390/diagnostics12122964.

[8] J. Corral-Acero *et al.*, "The 'Digital Twin' to enable the vision of precision cardiology," *Eur. Heart J.*, vol. 41, pp. 4556–4564, 2020, doi: 10.1093/eurheartj/ehaa159.

[9] G. Coorey *et al.*, "The health digital twin to tackle cardiovascular disease—A review of an emerging interdisciplinary field," *npj Digit. Med.*, vol. 5, Art. no. 126, 2022, doi: 10.1038/s41746-022-00640-7.

[10] S. A. Niederer *et al.*, "Scaling digital twins from the artisanal to the industrial," *Nat. Comput. Sci.*, vol. 1, pp. 313–320, 2021, doi: 10.1038/s43588-021-00072-5.

[11] F. Gu *et al.*, "Identification of digital twins to guide interpretable AI for diagnosis and prognosis in heart failure," *npj Digit. Med.*, vol. 8, Art. no. 110, 2025, doi: 10.1038/s41746-025-01501-9.

[12] J. Geddes *et al.*, "Digital twins for noninvasively measuring predictive markers of right heart failure," *npj Digit. Med.*, vol. 8, Art. no. 545, 2025, doi: 10.1038/s41746-025-01920-8.

[13] A. Bray *et al.*, "Pulse Physiology Engine: An open-source software platform for computational modeling of human medical simulation," *SN Compr. Clin. Med.*, vol. 1, pp. 362–377, 2019, doi: 10.1007/s42399-019-00053-w.

[14] K. P. Venkatesh, M. M. Raza, and J. C. Kvedar, "Health digital twins as tools for precision medicine: Considerations for computation, implementation, and regulation," *npj Digit. Med.*, vol. 5, Art. no. 150, 2022, doi: 10.1038/s41746-022-00694-7.

[15] A. Balasubramanyam, R. Ramesh, R. Sudheer, and P. B. Honnavalli, "Revolutionizing healthcare: A review unveiling the transformative power of digital twins," *IEEE Access*, vol. 12, pp. 69652–69676, 2024, doi: 10.1109/ACCESS.2024.3399744.

[16] E. Katsoulakis *et al.*, "Digital twins for health: A scoping review," *npj Digit. Med.*, vol. 7, Art. no. 77, 2024, doi: 10.1038/s41746-024-01073-0.

[17] A. Balasubramanyam *et al.*, "A cardiovascular modeling framework for enabling personalized healthcare: A digital twin approach," in *Proc. IEEE 13th Int. Conf. Healthcare Informatics (ICHI)*, 2025, pp. 478–489, doi: 10.1109/ICHI64645.2025.00062.

[18] Y. Shi, P. Lawford, and R. Hose, "Review of zero-D and 1-D models of blood flow in the cardiovascular system," *Biomed. Eng. Online*, vol. 10, Art. no. 33, 2011, doi: 10.1186/1475-925X-10-33.

[19] N. Westerhof, J.-W. Lankhaar, and B. E. Westerhof, "The arterial Windkessel," *Med. Biol. Eng. Comput.*, vol. 47, no. 2, pp. 131–141, 2009, doi: 10.1007/s11517-008-0359-2.

[20] H. Suga and K. Sagawa, "Instantaneous pressure-volume relationships and their ratio in the excised, supported canine left ventricle," *Circ. Res.*, vol. 35, no. 1, pp. 117–126, 1974, doi: 10.1161/01.RES.35.1.117.

[21] Kitware, Inc., "Pulse Physiology Engine: Circuit methodology." [Online]. Available: https://pulse.kitware.com/_circuit_methodology.html (accessed Oct. 12, 2026).

[22] A. Baird, M. McDaniel, S. A. White, N. Tatum, and L. Marin, "BioGears: A C++ library for whole body physiology simulations," *J. Open Source Softw.*, vol. 5, no. 56, Art. no. 2645, 2020, doi: 10.21105/joss.02645.

[23] T. Arts, T. Delhaas, P. Bovendeerd, X. Verbeek, and F. W. Prinzen, "Adaptation to mechanical load determines shape and properties of heart and circulation: The CircAdapt model," *Am. J. Physiol. Heart Circ. Physiol.*, vol. 288, no. 4, pp. H1943–H1954, 2005, doi: 10.1152/ajpheart.00444.2004.

[24] S. I. Chaudhry *et al.*, "Telemonitoring in patients with heart failure," *N. Engl. J. Med.*, vol. 363, no. 24, pp. 2301–2309, 2010, doi: 10.1056/NEJMoa1010029.

[25] M. K. Ong *et al.*, "Effectiveness of remote patient monitoring after discharge of hospitalized patients with heart failure: The Better Effectiveness After Transition–Heart Failure (BEAT-HF) randomized clinical trial," *JAMA Intern. Med.*, vol. 176, no. 3, pp. 310–318, 2016, doi: 10.1001/jamainternmed.2015.7712.

[26] F. Koehler *et al.*, "Efficacy of telemedical interventional management in patients with heart failure (TIM-HF2): A randomised, controlled, parallel-group, unmasked trial," *Lancet*, vol. 392, no. 10152, pp. 1047–1057, 2018, doi: 10.1016/S0140-6736(18)31880-4.

[27] J. Lindenfeld *et al.*, "Haemodynamic-guided management of heart failure (GUIDE-HF): A randomised controlled trial," *Lancet*, vol. 398, no. 10304, pp. 991–1001, 2021, doi: 10.1016/S0140-6736(21)01754-2.

[28] J. P. Boehmer *et al.*, "A multisensor algorithm predicts heart failure events in patients with implanted devices: Results from the MultiSENSE study," *JACC Heart Fail.*, vol. 5, no. 3, pp. 216–225, 2017, doi: 10.1016/j.jchf.2016.12.011.

[29] W. C. Levy *et al.*, "The Seattle Heart Failure Model: Prediction of survival in heart failure," *Circulation*, vol. 113, no. 11, pp. 1424–1433, 2006, doi: 10.1161/CIRCULATIONAHA.105.584102.

[30] S. J. Pocock *et al.*, "Predicting survival in heart failure: A risk score based on 39 372 patients from 30 studies," *Eur. Heart J.*, vol. 34, pp. 1404–1413, 2013, doi: 10.1093/eurheartj/ehs337.

[31] C. van Walraven *et al.*, "Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community," *CMAJ*, vol. 182, no. 6, pp. 551–557, 2010, doi: 10.1503/cmaj.091117.

[32] W. Ouwerkerk, A. A. Voors, and A. H. Zwinderman, "Factors influencing the predictive power of models for predicting mortality and/or heart failure hospitalization in patients with heart failure," *JACC Heart Fail.*, vol. 2, no. 5, pp. 429–436, 2014, doi: 10.1016/j.jchf.2014.04.006.

[33] B. J. Mortazavi *et al.*, "Analysis of machine learning techniques for heart failure readmissions," *Circ. Cardiovasc. Qual. Outcomes*, vol. 9, no. 6, pp. 629–640, 2016, doi: 10.1161/CIRCOUTCOMES.116.003039.

[34] D. Chicco and G. Jurman, "Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone," *BMC Med. Inform. Decis. Mak.*, vol. 20, Art. no. 16, 2020, doi: 10.1186/s12911-020-1023-5.

[35] M. Viceconti, F. Pappalardo, B. Rodriguez, M. Horner, J. Bischoff, and F. Musuamba Tshinanu, "In silico trials: Verification, validation and uncertainty quantification of predictive models used in the regulatory evaluation of biomedical products," *Methods*, vol. 185, pp. 120–127, 2021, doi: 10.1016/j.ymeth.2020.01.011.

[36] P. Pathmanathan and R. A. Gray, "Validation and trustworthiness of multiscale models of cardiac electrophysiology," *Front. Physiol.*, vol. 9, Art. no. 106, 2018, doi: 10.3389/fphys.2018.00106.

[37] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 11, pp. 2600–2606, 2018, doi: 10.1073/pnas.1708274114.

[38] A. E. W. Johnson *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Sci. Data*, vol. 10, Art. no. 1, 2023, doi: 10.1038/s41597-022-01899-x.

[39] J. L. Januzzi *et al.*, "NT-proBNP testing for diagnosis and short-term prognosis in acute destabilized heart failure: An international pooled analysis of 1256 patients: The International Collaborative of NT-proBNP Study," *Eur. Heart J.*, vol. 27, no. 3, pp. 330–337, 2006, doi: 10.1093/eurheartj/ehi631.

[40] A. L. Goldberger *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000, doi: 10.1161/01.CIR.101.23.e215.

[41] M. Kolakowski *et al.*, "Multimodal dataset of in-home physiological and inertial measurements from older heart failure patients," *Data*, vol. 11, Art. no. 106, 2026, doi: 10.3390/data11050106.

[42] Z. Zhang *et al.*, "Electronic healthcare records and external outcome data for hospitalized patients with heart failure," *Sci. Data*, vol. 8, Art. no. 46, 2021, doi: 10.1038/s41597-021-00835-9.

[43] J. Zhan *et al.*, "A multi-pathology ballistocardiogram dataset for cardiac function monitoring and arrhythmia assessment," figshare, 2025, doi: 10.6084/m9.figshare.28416896.

[44] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[45] Kitware, Inc., "Pulse Physiology Engine: Cardiovascular methodology." [Online]. Available: https://pulse.kitware.com/_cardiovascular_methodology.html (accessed Oct. 6, 2026).

[46] N. Stergiopulos, J.-J. Meister, and N. Westerhof, "Determinants of stroke volume and systolic and diastolic aortic pressure," *Am. J. Physiol. Heart Circ. Physiol.*, vol. 270, no. 6, pp. H2050–H2059, 1996, doi: 10.1152/ajpheart.1996.270.6.H2050.

[47] Kitware, Inc., "Pulse Physiology Engine, release 4.3.1, source file `CardiovascularModel.cpp` (functions `BeginDriverCycle` and `ChronicHeartFailure`)." [Online]. Available: https://gitlab.kitware.com/physiology/engine/-/blob/REL_4_3_1/src/cpp/engine/common/system/physiology/CardiovascularModel.cpp (accessed Oct. 12, 2026).

[48] Kitware, Inc., "Pulse Physiology Engine, release 4.3.1, source file `NervousModel.cpp` (function `BaroreceptorFeedback`)." [Online]. Available: https://gitlab.kitware.com/physiology/engine/-/blob/REL_4_3_1/src/cpp/engine/common/system/physiology/NervousModel.cpp (accessed Oct. 13, 2026).

[49] Royal College of Physicians, "National Early Warning Score (NEWS) 2: Standardising the assessment of acute-illness severity in the NHS," London, U.K., 2017.

[50] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[51] J.-L. Vincent and D. De Backer, "Circulatory shock," *N. Engl. J. Med.*, vol. 369, no. 18, pp. 1726–1734, 2013, doi: 10.1056/NEJMra1208943.

[52] A. Nohria, E. Lewis, and L. W. Stevenson, "Medical management of advanced heart failure," *JAMA*, vol. 287, no. 5, pp. 628–640, 2002, doi: 10.1001/jama.287.5.628.

[53] D. A. Baran *et al.*, "SCAI clinical expert consensus statement on the classification of cardiogenic shock," *Catheter. Cardiovasc. Interv.*, vol. 94, pp. 29–37, 2019, doi: 10.1002/ccd.28329.

[54] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.